

AI – Powered Autonomous Search And Rescue Drone

Presented by Team Garuda

ABSTRACT

GARUDA is an AI-powered autonomous drone system designed to support **search-and-rescue operations in disaster-affected and hazardous environments**. During disasters such as floods, earthquakes, landslides, fires, and structural collapses, rescue teams often face dangerous conditions, limited visibility, inaccessible locations, and delays in locating survivors. GARUDA addresses these challenges by providing an aerial platform capable of surveying affected areas, detecting survivors and identifying environmental hazards before responders enter the danger zone.

The proposed system integrates **RGB and thermal cameras, GPS, IMU, and onboard sensors** to collect visual, thermal, positional, and environmental information. The captured data is processed using **on-device/edge AI**, enabling real-time detection of people and hazards without depending entirely on cloud connectivity. Computer vision techniques are used to identify survivors and recognize hazards such as **fire, floodwaters, damaged structures, debris, landslides, and exposed dangerous areas**. Thermal sensing complements RGB vision by helping identify human heat signatures in conditions where visibility may be poor.

GARUDA is designed to perform **autonomous area surveying** using predefined waypoints or adaptive navigation. Detected survivors and hazards are **geo-tagged and placed on a real-time disaster map**, providing rescue teams with a clearer understanding of the affected region. The system also assigns different risk levels to mapped areas and uses **A* path planning** to generate a **risk-aware recommended access route**, helping responders identify routes based on detected hazards and terrain conditions rather than simply choosing the shortest path.

A **command-center interface** can receive important information such as drone location, survivor detections, hazard locations, mapped areas, alerts, and recommended access routes. This allows emergency response teams to monitor the situation remotely and prioritize their operations using updated aerial intelligence. Local processing also allows critical detection and decision-making functions to continue in areas with weak or unavailable network connectivity, with information synchronized when communication becomes available.

The proposed solution follows a modular architecture combining **multi-modal sensing, computer vision, edge AI, autonomous navigation, mapping, risk assessment, and emergency communication**. By providing early information about survivors, hazards, and accessible areas, GARUDA aims to **reduce responder exposure to dangerous environments, reduce victim discovery time, and improve situational awareness during disaster response**. The system can further be extended toward **multi-drone coordination, advanced sensor integration, improved autonomous navigation, and integration with larger disaster-management networks**.

Keywords: Autonomous Drone, Search and Rescue, Edge AI, Thermal Imaging, Computer Vision, Survivor Detection, Hazard Detection, Disaster Mapping, A* Path Planning, Risk-Aware Navigation, Emergency Response.

INTRODUCTION

Natural disasters and large-scale emergencies can create situations where every second matters. Floods, earthquakes, landslides, fires, building collapses, and other disasters can affect large areas and make normal rescue operations extremely difficult. Roads may be blocked, buildings may become unstable, visibility may be poor, and dangerous conditions such as fire, floodwater, debris, or exposed electrical lines may put rescue personnel at risk. In such situations, the first challenge is often not only reaching the affected area, but also **knowing where survivors are located and understanding the risks present before entering the area.**

Traditional search-and-rescue operations mainly depend on teams physically entering the affected region and searching for victims. Although trained rescue personnel are essential, sending people directly into an unknown disaster environment can expose them to additional risks. Searching large areas manually can also require significant time and manpower. In locations that are inaccessible by road or surrounded by debris, water, unstable structures, or other hazards, ground-based inspection may be slow or difficult. This creates a need for technologies that can provide rescue teams with information about the situation **before they enter the danger zone.**

Unmanned aerial vehicles, commonly known as drones, provide an effective way to obtain an aerial view of disaster-affected regions. A drone can cover areas that may be difficult for humans to reach and can collect information from multiple locations in a relatively short period. With cameras and other sensors mounted on the drone, it can observe the environment from above and provide useful information to emergency response teams. However, simply using a camera-equipped drone is not enough for an intelligent search-and-rescue system. The large amount of visual and sensor data collected during a mission must be processed quickly to identify important events such as the presence of a survivor or a dangerous environmental condition.

This is where **Artificial Intelligence (AI), computer vision, and edge computing** become important. Instead of depending completely on a remote server or cloud platform, an AI-enabled drone can process critical sensor information directly on the onboard computing system. This approach can reduce the delay between capturing information and detecting an important event. It can also be useful in disaster areas where communication networks may be damaged, overloaded, or completely unavailable. For a search-and-rescue mission, the ability to continue critical detection functions even with limited connectivity can be particularly valuable.

1.1 Background of the Project

The **GARUDA** project is proposed as an AI-powered autonomous drone system for assisting search-and-rescue teams during disaster situations. The main idea is to allow the drone to enter and inspect potentially dangerous areas first, collect information about survivors and hazards, and provide that information to rescue personnel. The system combines aerial mobility with **RGB imaging, thermal sensing, GPS, IMU, onboard AI, autonomous navigation, mapping, and risk assessment.**

The use of both RGB and thermal cameras provides complementary information. An RGB camera can capture normal visual information about the surrounding environment and can be used for detecting people, debris, structures, fire-related visual features, and other objects. A thermal camera can provide heat-based information and can help identify human heat signatures when normal visual detection becomes difficult. Combining these sensing methods gives the system a broader understanding of the environment than relying on a single camera type.

GARUDA also focuses on **local or onboard AI processing**. Sensor information captured by the drone can be processed on the drone itself to detect survivors and hazards. This reduces the need to continuously send raw sensor data to a remote system for processing. Instead, important information can be extracted locally and then transmitted to the command center. This approach is particularly useful for disaster-response scenarios where network availability cannot always be assumed.

The drone is also designed to support autonomous surveying of affected areas. Instead of requiring an operator to manually control every movement, the proposed system can use planned waypoints or survey patterns to cover a defined region. The drone can continuously capture sensor information while moving through the area and process the information to identify important findings.

1.2 Need for the Proposed System

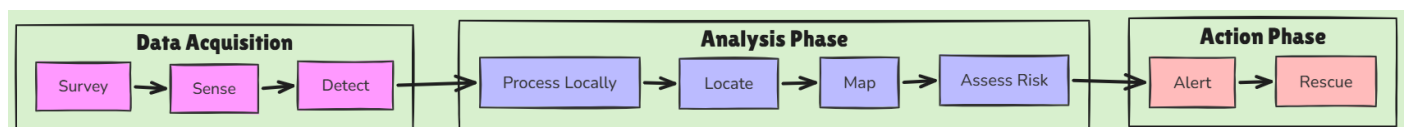
One of the major difficulties during disaster response is the lack of **real-time situational awareness**. Rescue teams may know that a particular region has been affected, but they may not have accurate information about the current condition of that region. For example, a road that was previously accessible may now be blocked by debris or floodwater. A building that appears accessible from a distance may have structural damage. A fire or other hazard may also prevent rescuers from safely approaching a particular location.

GARUDA is intended to provide this information through aerial observation and AI-based analysis. When a survivor or hazard is detected, its location can be recorded using the drone's positioning information. These geo-tagged detections can then be displayed on a disaster map. Instead of receiving only a video feed, rescue teams can receive more structured information such as **where a survivor was detected, where a hazard exists, and which areas may require additional attention**.

Another important requirement is that rescue teams should not simply receive information about hazards; they should also be able to use that information for planning their movement. For this reason, GARUDA includes the concept of **risk-aware route planning**. Areas identified as dangerous can be assigned higher risk costs, while relatively clear areas can have lower costs. An **A* path-planning algorithm** can then be used to calculate a recommended route based on the mapped environment and risk values. The purpose is not to claim that the route is guaranteed to be safe, but to provide rescue teams with an informed route recommendation based on the available drone observations.

1.3 GARUDA's Approach

The overall approach of GARUDA can be understood as a sequence of connected operations:



First, the drone takes off and initializes its navigation and sensing systems. It then surveys the assigned disaster-affected region using autonomous movement. During the survey, RGB and thermal cameras continuously capture information about the environment.

The captured information is processed using onboard AI and computer vision techniques. The system attempts to identify survivors and relevant hazards. When an important detection is made, the corresponding location is geo-tagged and added to the disaster map. The detected hazard can also contribute to the risk level assigned to that region.

The updated map can then be used for generating risk-aware route recommendations. Important information is transmitted to the command center, where rescue personnel can monitor the drone's position, detected survivors, hazards, and other mission information. Based on this information, response teams can make their operational decisions.

This approach changes the role of the drone from being only an **aerial camera** into an intelligent information-gathering system. Instead of requiring rescue personnel to continuously interpret a large amount of raw video, the system is designed to identify and highlight important information that can support emergency response.

1.4 Importance of Autonomous and Offline Operation

Disaster environments are often unpredictable. Communication infrastructure can be damaged by earthquakes, floods, fires, or structural failures. Even when communication is available, bandwidth may be limited when multiple emergency teams are operating in the same region. Therefore, a rescue-support system should not depend entirely on continuous cloud connectivity for its most important functions.

GARUDA addresses this requirement through edge AI and local processing. Critical detection tasks can be performed on the drone, allowing the system to continue its primary perception functions even when network connectivity is weak or unavailable. When communication is available, important detections and mission information can be sent to the command center. This creates a more resilient operating model in which the drone can continue collecting and processing information locally and synchronize important information when connectivity becomes available.

Autonomous navigation is another important part of the system. A disaster area may contain obstacles and conditions that make direct manual operation difficult. GARUDA therefore considers GPS-based navigation when GPS is available and sensor-based localization approaches such as **SLAM and IMU-assisted navigation** for situations where GPS availability may be limited. Obstacle detection and avoidance can also contribute to safer drone movement.

1.5 Overall Purpose of GARUDA

The primary purpose of GARUDA is to provide rescue teams with **early, actionable information about survivors and hazards**. The project does not aim to replace human rescue personnel. Instead, it is designed to act as an intelligent aerial support system that can inspect dangerous or difficult-to-access areas before responders enter them.

The system brings together several technologies into one workflow: **multi-modal sensing, AI-based detection, edge computing, autonomous navigation, geo-tagging, disaster mapping, risk assessment, route planning, and command-center monitoring**. Each component contributes to the overall objective of improving the information available to rescue teams.

The central vision of the project can be summarized as:

“Let the drone enter the danger zone before the rescuer does.”

By providing an aerial view of the disaster environment and identifying important findings automatically, GARUDA aims to support faster and safer decision-making during emergency response. The system can also be extended in the future to support multiple coordinated drones, additional sensors, improved autonomous navigation, and integration with wider disaster-management infrastructure.

Overall, GARUDA represents a practical application of AI, robotics, computer vision, and autonomous systems to one of the most challenging real-world problems: **finding people and understanding hazards quickly when the environment itself makes human access dangerous**.

HARDWARE REQUIREMENT

The hardware of the **GARUDA** system forms the physical foundation of the entire search-and-rescue platform. Since the proposed system is intended to operate in disaster-affected areas, the hardware must support stable flight, continuous sensing, onboard AI processing, communication, navigation, and data collection while keeping the overall drone lightweight and power-efficient.

Unlike a conventional camera drone, GARUDA requires additional hardware for **survivor detection, thermal sensing, hazard identification, autonomous navigation, geo-tagging, and edge AI processing**. The hardware architecture is therefore designed in a modular manner, allowing the existing drone platform to be progressively enhanced with additional sensors and computing components.

The major hardware elements include the drone frame, propulsion system, flight controller, battery and power system, RGB and thermal cameras, GPS, IMU, onboard edge-computing unit, communication modules, storage, and a ground-based command-center computer.

The proposed hardware architecture follows the general flow:



The following sections describe the hardware requirements of GARUDA in detail.

2.1 Drone Airframe

The **drone airframe** is the main physical structure on which all the other components are mounted. GARUDA is designed around a multirotor/quadcopter platform because a multirotor can take off and land vertically and can hover over a particular location while collecting information.

This is particularly useful in disaster environments where roads may be blocked or ground access may be difficult.

The airframe needs to provide sufficient space and payload capacity for:

- RGB camera
- Thermal camera
- Flight controller
- GPS and other navigation sensors
- Edge AI computing unit
- Communication equipment
- Battery and power distribution system
- Optional obstacle-detection sensors

The frame should be lightweight but mechanically strong enough to withstand normal vibration and the additional load of the sensing and computing equipment.

2.2 Important requirements

The selected frame should provide:

- Adequate payload capacity
- Good mechanical stability
- Low vibration during flight
- Proper mounting locations for cameras and sensors
- Space for onboard computing hardware
- Protection for sensitive electronics
- Easy access for maintenance and component replacement

The existing drone platform can be used as the base platform, with the additional GARUDA components integrated progressively.

2.3 Propulsion System

The propulsion system is responsible for generating the thrust required to keep the drone airborne and control its movement.

For a quadcopter configuration, the propulsion system generally consists of:

- Four brushless motors
- Four Electronic Speed Controllers (ESCs)
- Propellers
- Battery
- Power distribution system

The propulsion system must be selected according to the total weight of the drone after adding the cameras, computing unit, communication hardware and other sensors.

2.3.1 Brushless Motors

Brushless DC motors are used to generate the required thrust.

The motor selection should consider:

- Required thrust
- Total drone weight
- Propeller compatibility
- Power consumption
- Operating efficiency
- Reliability

Since GARUDA will carry additional payload compared with a basic recreational drone, the motors must provide sufficient thrust without excessively increasing power consumption.

2.3.2 Electronic Speed Controllers

ESCs control the speed of the individual motors according to commands received from the flight controller.

The ESCs should be compatible with the selected motors and battery system. Reliable ESC operation is important because an unexpected motor-control failure can directly affect flight stability.

2.4 Flight Controller

The **flight controller** is responsible for the basic operation and stabilization of the drone.

It continuously receives information from sensors such as the IMU and GPS and uses this information to control the motors.

The flight controller handles functions such as:

- Stabilization
- Attitude control
- Altitude control
- Motor control
- Take-off and landing
- Position control
- Waypoint navigation
- Failsafe functions

A suitable autopilot firmware such as **ArduPilot** can be considered for the flight-control layer.

The flight controller and the onboard AI computer have different responsibilities. The flight controller manages the **low-level flight and safety functions**, while the edge computer performs **high-level AI, perception, mapping and mission logic**.

This separation makes the architecture more modular and safer because AI processing does not need to directly replace the fundamental flight-control system.

2.5 Battery and Power Supply

The battery is one of the most important hardware components because every subsystem of GARUDA ultimately depends on the available electrical power.

A rechargeable **Li-Po battery** can be used to supply power to the propulsion system and onboard electronics.

The battery needs to provide sufficient energy for:

- Motors
- Flight controller
- Cameras
- Edge AI computer

- GPS
- Communication system
- Additional sensors

The actual battery capacity should be selected only after calculating the final weight and power consumption of the complete prototype.

2.5.1 Power-related requirements

The power system should provide:

- Stable power delivery
- Adequate current capacity
- Battery monitoring
- Protection against excessive current
- Suitable voltage regulation
- Separate regulated supplies for sensitive electronics where necessary

A major design challenge is the balance between **computational capability and flight endurance**. A more powerful onboard computer can improve AI processing capability but can also increase power consumption. Therefore, the hardware selection must consider both AI performance and battery life.

2.6 Power Distribution and Regulation

A **Power Distribution Board (PDB)** distributes electrical power from the main battery to different components of the drone.

The propulsion system and electronic components may require different voltage levels. Therefore, suitable voltage regulators or power modules may be required for the onboard electronics.

The power system should be designed so that:

- Motors receive the required power.
- Flight-control electronics receive stable power.
- Cameras receive their required voltage.
- The edge computer receives regulated power.
- Communication modules receive stable power.
- Battery status can be monitored.

Proper power management is especially important for GARUDA because onboard AI processing and thermal imaging introduce additional electrical loads.

2.7 RGB Camera

The **RGB camera** is the primary visual sensing component of GARUDA.

It provides normal visible-spectrum images of the environment and acts as one of the major data sources for the AI system.

The RGB camera can be used for:

- Survivor detection
- Human presence detection
- Fire detection
- Debris detection
- Structural damage assessment
- Visual mapping
- Scene understanding
- Navigation support

The camera should provide sufficient image quality for computer-vision processing while maintaining reasonable size, weight and power consumption.

During flight, the RGB camera continuously captures visual information. This information is transferred to the onboard computing system, where AI models can analyze the images and identify objects or situations of interest.

2.8 Thermal Camera

The **thermal camera** provides information that cannot always be obtained from a normal RGB camera.

Instead of relying only on visible appearance, thermal imaging provides information based on heat signatures. This makes it particularly useful for search-and-rescue operations.

For example, a survivor may be difficult to identify using a normal camera because of:

- Darkness
- Smoke
- Debris
- Poor visibility
- Partial obstruction
- Low-light conditions

Thermal sensing can provide an additional source of information in such situations.

GARUDA therefore uses RGB and thermal sensing as complementary modalities.

RGB + Thermal sensing

The two cameras provide different types of information:

In the form of tabular form.

RGB Camera	Thermal Camera
Visible image information	Heat-signature information
Useful in normal lighting	Useful in low-visibility conditions
Detects visual characteristics	Detects thermal characteristics
Useful for scene understanding	Useful for identifying possible human heat signatures

The combination of both sensing methods can improve the robustness of survivor-search operations compared with relying on a single visual sensor.

2.9 GPS/GNSS Module

A **GPS/GNSS module** provides the geographical location of the drone.

Location information is essential to GARUDA because detecting a survivor is only useful to rescue teams if the system can also communicate **where that survivor was detected**.

GPS information can be used for:

- Drone positioning
- Autonomous waypoint navigation
- Geo-tagging survivors
- Geo-tagging hazards
- Recording flight paths
- Disaster mapping
- Reporting locations to the command center

For example, if GARUDA detects a person during its survey, the detection can be associated with the drone's current coordinates and added to the disaster map.

However, GPS cannot be assumed to work reliably in every environment. Disaster zones containing large structures or areas where satellite signals are degraded may require alternative localization methods.

2.10 Inertial Measurement Unit (IMU)

The **IMU** provides information about the motion and orientation of the drone.

It typically provides measurements related to:

- Acceleration
- Angular velocity
- Orientation

The IMU is important for flight stabilization and navigation.

GARUDA can combine IMU information with GPS and camera-based information to obtain a more reliable estimate of the drone's position and movement.

This becomes particularly important when GPS information is temporarily unavailable or unreliable.

2.11 Onboard Edge AI Computing Unit

One of the most important additions to the GARUDA platform is the **onboard edge AI computer**.

A conventional drone may simply capture images and send them to another computer for analysis. GARUDA instead proposes performing important AI processing directly on the drone.

The edge computer is responsible for computational tasks such as:

- Image processing
- Survivor detection
- Hazard detection
- RGB and thermal data processing
- Sensor fusion
- Mapping
- Localization
- SLAM
- Risk assessment
- A* route planning
- Mission-level decision support

This approach reduces dependence on continuous internet connectivity.

2.11.1 Proposed Edge AI Platform

A **Qualcomm Flight RB5 / Qualcomm Robotics RB5-class platform** can be considered as a proposed edge-computing option for the system.

The exact platform should ultimately be selected based on:

- AI processing capability
- Camera interfaces
- Power consumption
- Weight
- Available memory
- Storage
- Connectivity
- ROS 2 compatibility

- Thermal requirements
- Physical integration with the drone

The RB5 platform should be described as a **proposed/reference hardware platform** unless the specific hardware has been acquired and integrated into the prototype.

2.12 Communication and Telemetry Hardware

GARUDA requires a communication link between the drone and the ground command center.

The communication system can be based on suitable technologies such as:

- Radio telemetry
- Wi-Fi
- 4G/5G
- Other appropriate wireless communication technologies

The exact communication technology will depend on the deployment environment.

The communication system can transmit information such as:

- Drone location
- Battery status
- Flight status
- Survivor detections
- Hazard detections
- Coordinates
- Alerts
- Mission status
- Mapping information

However, GARUDA is not designed to depend entirely on network connectivity. Critical detection and processing functions are intended to operate locally on the drone.

This is important because disaster areas may experience damaged communication infrastructure or weak network coverage.

2.13 Onboard Storage

Local storage is required for storing information generated during a mission.

The drone may need to store:

- RGB images
- Thermal images

- Detection results
- GPS coordinates
- Sensor readings
- Flight logs
- Mission information
- Disaster maps
- System diagnostic data

Local storage also supports offline operation. If communication is temporarily unavailable, important information can remain stored on the drone and can be synchronized with the command center when communication is restored.

The storage capacity should therefore be selected according to the expected mission duration and data-generation rate.

2.14 Optional Microphone Array

A microphone array can be included as an additional sensing module.

The purpose of this component is to provide an **acoustic sensing capability** that complements the visual and thermal sensors.

The system could potentially use audio information to identify:

- Human voices
- Calls for help
- Crying or shouting
- Other relevant sounds

This is an optional component rather than a mandatory requirement for the initial prototype.

Its inclusion would depend on the available payload, power budget and the effectiveness of acoustic detection in the intended environment.

2.15 Obstacle Detection Sensors

Autonomous operation requires the drone to be aware of obstacles in its surroundings.

Depending on the final implementation, GARUDA can use additional distance or depth sensors such as:

- LiDAR
- Depth camera
- Ultrasonic sensor
- Other range sensors

These sensors can assist in identifying obstacles such as:

- Buildings
- Trees
- Poles
- Debris
- Walls
- Other structures

The exact sensor should be selected based on the required detection range, weight, power consumption and environmental conditions.

These sensors are considered an **optional/additional hardware layer** and should not be presented as already implemented unless they are actually integrated into the prototype.

2.16 Ground Station / Command-Center Computer

The drone requires a ground-based computing system through which rescue personnel can monitor the mission.

This can be a laptop, workstation or other suitable computer.

The command-center system can display:

- Drone position
- Drone status
- Survivor locations
- Hazard locations
- Disaster map
- Alerts
- Flight information
- Recommended risk-aware access routes
- Mission progress

The ground station does not necessarily need to perform all AI processing because the critical perception tasks are intended to run on the onboard edge computer.

This allows the command center to focus primarily on **monitoring, visualization and human decision-making**.

2.17 Camera and Sensor Mounting

Proper mounting of sensors is an important part of the hardware design.

The RGB and thermal cameras should be positioned so that their field of view covers the required area beneath and around the drone.

The mounting system should:

- Minimize vibration
- Maintain camera orientation
- Protect the cameras from accidental impact
- Allow easy replacement
- Avoid obstructing the propellers
- Maintain the required field of view

If RGB and thermal cameras are mounted separately, their relative orientation should also be considered so that information from both sensors can be appropriately associated during processing.

2.18 Payload and Weight Considerations

One of the major hardware challenges in GARUDA is **payload management**.

Every additional component increases the total weight of the drone. This affects:

- Required motor thrust
- Battery consumption
- Flight duration
- Stability
- Maneuverability

Therefore, the hardware should not simply be selected based on computational or sensing capability. Each component should also be evaluated based on its **weight, power requirement and physical dimensions**.

The final system should maintain an appropriate balance between:

Payload + Computing Capability + Sensing Capability + Flight Endurance

This is particularly important for the edge AI computer and thermal camera because both provide significant functionality but also add additional weight and power requirements.

2.19 Environmental Protection

GARUDA is intended for disaster-response scenarios, where the environment may be considerably harsher than a normal drone operating environment.

Potential conditions include:

- Rain
- Dust
- Smoke
- High temperatures
- Strong winds

- Debris
- Moisture
- Unstable structures

Therefore, appropriate protection should be considered for sensitive electronic components.

Depending on the final design, protective measures may include:

- Protective enclosures
- Sealed connectors
- Proper cable management
- Vibration isolation
- Thermal management
- Camera protection

The drone should not be assumed to operate safely in every extreme condition. Environmental operating limits should be defined and validated through controlled testing.

2.20 Thermal Management

The edge AI computer can generate heat during continuous AI inference and image processing.

Therefore, thermal management must be considered when installing the computing unit inside the drone.

Possible approaches include:

- Heat sinks
- Passive cooling
- Controlled airflow
- Proper enclosure design
- Thermal monitoring

The selected solution should provide sufficient cooling without significantly increasing the weight or power consumption of the system.

2.21 Hardware Safety and Reliability

Since GARUDA is intended to operate in potentially dangerous environments, hardware reliability is important.

The system should incorporate appropriate safety mechanisms at both the flight-controller and hardware levels.

Important considerations include:

- Battery monitoring
- Pre-flight hardware checks

- Reliable motor and ESC connections
- Secure mounting of payload
- Communication-loss handling
- Low-battery handling
- Controlled landing/return procedures
- Protection against loose wiring
- Reducing vibration
- Proper maintenance before flight

The AI system should not be treated as a replacement for the drone's fundamental flight-safety mechanisms.

SOFTWARE REQUIREMENTS

The **AI-Powered Autonomous Search and Rescue Drone** requires software for autonomous flight control, AI-based human detection, image processing, navigation, communication, and monitoring. The following software components are specifically required for the proposed project.

4.1. Operating System

Ubuntu Linux / Windows 10 or later

The operating system provides the environment required to develop, configure, test, and run the drone's AI and control software.

4.2. Python

Python is used as the primary programming language for the AI and computer vision components of the project. It is used for:

- Processing camera data.
- Running the human detection model.
- Processing GPS information.
- Implementing search and rescue logic.
- Generating alerts when a person is detected.
- Integrating the AI system with the drone-control software.

4.3. YOLO

YOLO (You Only Look Once) is used for real-time human detection from the drone's camera feed.

It is responsible for:

- Detecting humans in captured images and video.
- Identifying the position of detected persons within the image.
- Providing a confidence value for each detection.
- Supporting real-time detection during autonomous flight.

4.4. PyTorch

PyTorch is used as the deep-learning framework for running the YOLO-based detection model.

It provides the required environment for:

- Loading the trained AI model.
- Running inference on camera frames.
- Processing neural-network operations.
- Evaluating human-detection results.

4.5. OpenCV

OpenCV (Open Source Computer Vision Library) is used for processing the video and images captured by the drone camera.

It performs:

- Video-frame extraction.
- Image resizing and preprocessing.
- Image enhancement.
- Processing of live camera feeds.
- Displaying and storing detected-person images.
- Integration of camera data with the AI detection model.

4.6. ArduPilot

ArduPilot is used as the flight-control software for the drone.

It manages:

- Autonomous take-off and landing.
- Drone movement and stabilization.
- Waypoint-based navigation.
- Altitude control.
- Autonomous search missions.
- Return-to-home operation.
- Flight-status monitoring.

4.7. MAVLink

MAVLink (Micro Air Vehicle Link) is used as the communication protocol between the flight controller and the software running on the onboard computer.

It is used to exchange:

- GPS coordinates.
- Altitude.

- Speed.
- Battery status.
- Flight mode.
- Telemetry data.
- Mission commands.
-

4.8. MAVSDK

MAVSDK is used to connect the Python application with the drone's flight controller.

It allows the project software to:

- Send navigation commands.
- Start and stop missions.
- Receive GPS and telemetry information.
- Monitor the drone's flight status.
- Communicate with ArduPilot through MAVLink.

4.9. GPS/GNSS Software Integration

GPS/GNSS integration is required for autonomous navigation and location identification.

The software uses GPS data to:

- Determine the drone's current location.
- Navigate through predefined search areas.
- Follow planned waypoints.
- Record the approximate location where a person is detected.
- Provide location information to the rescue team.

4.10. Mission Planner

Mission Planner is used as the ground-control software for configuring and monitoring the drone.

It provides:

- Mission planning.
- Waypoint configuration.
- Real-time drone location.
- Flight-path monitoring.
- Telemetry monitoring.

- Battery-status monitoring.
- Manual control when required.
- Mission-status information.

4.11. SQLite Database

SQLite is used to store the important information generated during the search and rescue mission.

The database stores:

- Mission information.
- Detection records.
- GPS coordinates.
- Detection confidence.
- Detection time.
- Records of identified persons.

4.12. Visual Studio Code

Visual Studio Code (VS Code) is used as the primary development environment for the project.

It is used to:

- Develop Python programs.
- Integrate YOLO, PyTorch, and OpenCV.
- Develop drone-control programs.
- Test individual software modules.
- Debug and maintain the project source code.

METHODOLOGY

The methodology of **GARUDA** is designed to convert the drone from a conventional aerial platform into an intelligent search-and-rescue system. The proposed workflow combines *autonomous area surveying, RGB and thermal sensing, on-device AI, survivor and hazard detection, geo-tagging, disaster mapping, risk assessment, A path planning, and communication with the command center**.

The system follows a continuous process in which the drone first surveys the disaster-affected area, collects sensor data, processes the information locally, identifies survivors and hazards, records their locations, updates the disaster map, and sends important information to the command center.

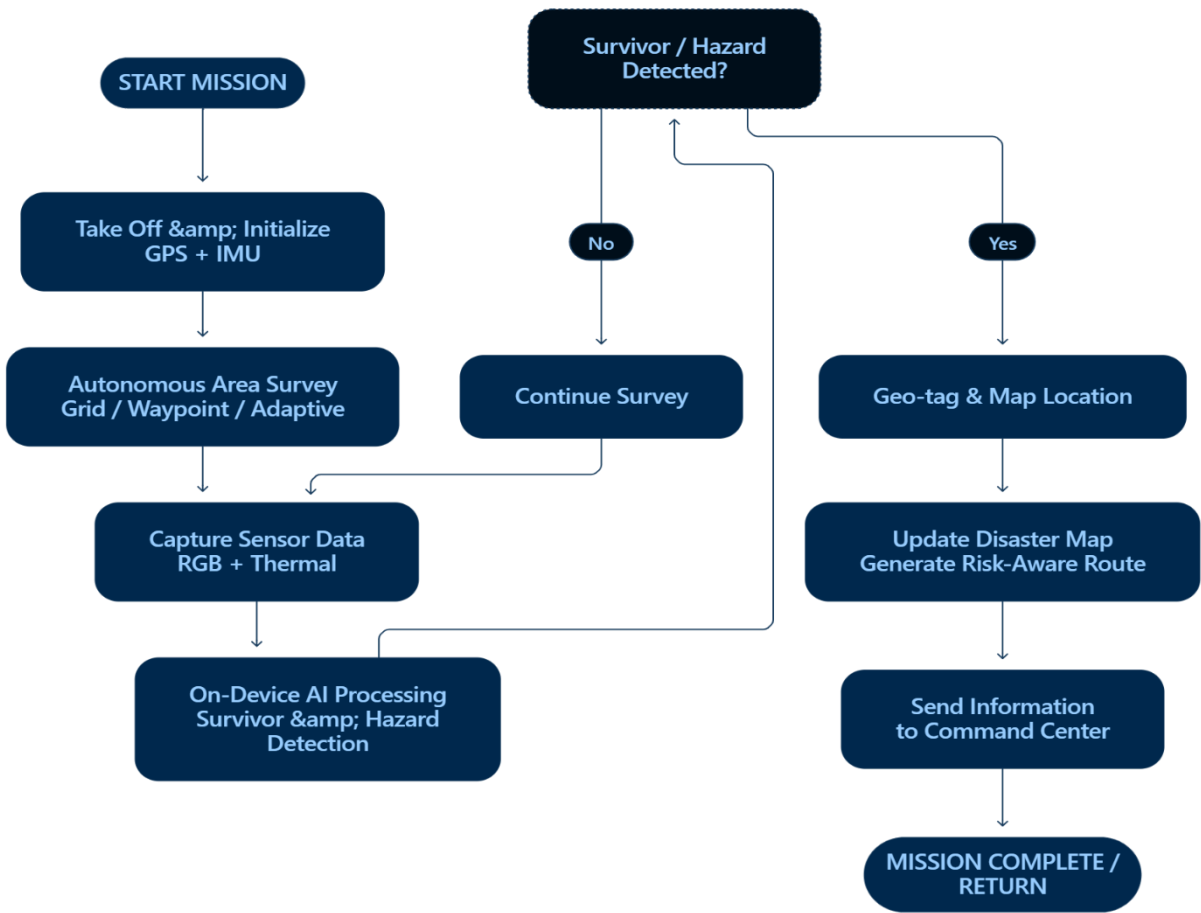
The overall methodology can be represented as:



The methodology is divided into a sequence of interconnected stages, as described below.

5.1. Overall System Workflow

Methodology Flow Diagram



This workflow represents the proposed GARUDA methodology and does not introduce additional functions outside the project's defined system.

5.1.1. Stage 1 – Start Mission

The workflow begins when the GARUDA drone is assigned a search-and-rescue mission.

The mission area is identified as the region that needs to be surveyed. The drone is prepared for autonomous operation and the required onboard systems are initialized.

At this stage, the system prepares:

- Flight controller
- GPS
- IMU
- RGB camera
- Thermal camera
- Onboard AI processing
- Communication with the command center

The purpose of this stage is to ensure that the drone is ready to begin the aerial survey.

5.1.2. Stage 2 – Take Off and Initialize

After the mission begins, the drone takes off and initializes its navigation sensors.

The **GPS** provides the geographical position of the drone, while the **IMU** provides information about its movement and orientation.

The initial position and sensor information provide the foundation for:

- Autonomous navigation
- Location tracking
- Geo-tagging
- Disaster mapping

The drone then proceeds toward the area selected for surveying.

5.1.3 Stage 3 – Autonomous Area Survey

Once the drone reaches the required area, it begins an autonomous survey.

The survey can follow a predefined **grid or waypoint-based pattern**, allowing the drone to systematically cover the target region.

The purpose of the survey is to collect sufficient information about the disaster-affected area without requiring a person to manually control every movement of the drone.

During the survey, the drone continuously moves through the target area while collecting information from its cameras and other onboard sensors.

The survey continues until the required area has been sufficiently covered or the mission is completed.

5.1.4 Stage 4 – Sensor Data Acquisition

While surveying, GARUDA continuously collects information using its sensing system.

The primary sensing modalities are:

RGB Camera

The RGB camera captures normal visual information from the environment.

It provides information useful for identifying:

- People
- Fire
- Debris
- Damaged surroundings
- Other visually identifiable hazards

Thermal Camera

The thermal camera provides heat-based information.

It is particularly useful for detecting possible human heat signatures in situations where normal visual detection may be difficult.

GPS and IMU

GPS and IMU provide positional and motion information that can later be associated with detected objects.

The information from these sensors forms the input for the onboard processing stage.

5.1.5 Stage 5 – On-Device AI Processing

The captured sensor information is processed locally on the drone using the proposed **edge/on-device AI system**.

This is an important part of the GARUDA methodology.

Instead of sending every captured image to a remote cloud system for analysis, the drone performs the critical detection process locally.

The onboard processing pipeline analyzes the incoming sensor information and identifies potential:

- Survivors
- Human presence
- Fire
- Floodwaters
- Debris
- Damaged structures

- Other defined disaster hazards

The RGB and thermal information can be used together to improve the overall understanding of the environment.

Local processing also helps reduce dependence on continuous internet connectivity.

5.1.6 Stage 6 – Survivor and Hazard Detection

After receiving the sensor data, the AI system determines whether a survivor or relevant hazard is present.

The detection stage has two primary objectives:

Survivor Detection

The system analyzes the RGB and thermal information to identify possible survivors or signs of human presence.

When a survivor is detected, the system records the detection and associates it with the drone's current location.

Hazard Detection

The system also identifies relevant environmental hazards.

The project's defined hazards include:

- Fire
- Floodwaters
- Damaged structures
- Exposed electrical lines
- Debris
- Landslides
- Chemical leaks

The detected hazards are treated as important information for understanding the condition of the disaster area and planning responder access.

5.1.7. Stage 7 – Detection Decision

After AI processing, the system checks:

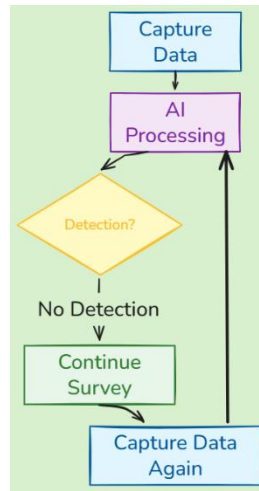
“Was a survivor or hazard detected?”

There are two possible paths.

If no survivor or hazard is detected

The drone continues its survey.

It returns to the sensor-data acquisition stage and continues collecting information from the next area.



This loop allows GARUDA to continuously search the target area rather than stopping after every frame or observation.

If a survivor or hazard is detected

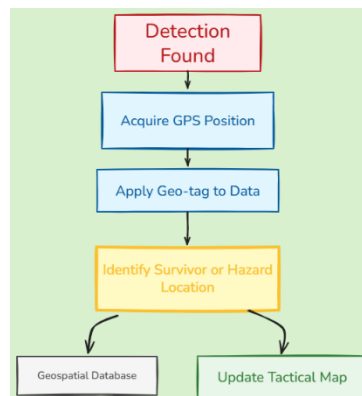
The system proceeds to the geo-tagging and mapping stage.

5.1.8. Stage 8 – Geo-Tagging the Detection

When a survivor or hazard is detected, GARUDA records its geographical location.

The detected object is associated with the drone's location information obtained through the navigation system.

For example:



This converts a simple AI detection into actionable location information.

For rescue teams, knowing that a survivor exists is important, but knowing **where the survivor was detected** is much more useful for planning the response.

The same process is applied to detected hazards.

5.1.9. Stage 9 – Disaster Map Update

The geo-tagged detections are added to the **disaster map**.

The map provides a visual representation of the surveyed area and the important information detected by GARUDA.

The map can contain:

- Drone survey information
- Survivor locations
- Hazard locations
- Relevant mapped areas
- Risk information

As the drone continues its survey, newly detected information can be added to the map.

This gradually provides the rescue team with a clearer picture of the affected region.

5.1.10. Stage 10 – Risk Assessment

After hazards are identified and mapped, GARUDA assigns different levels of risk to the affected areas.

The purpose of risk assessment is to distinguish between relatively accessible areas and areas where hazards may make movement more dangerous.

For example, the project can assign different risk costs to different mapped conditions:

Area Condition	Example Risk Cost
Clear/normal area	Low
Debris	Medium
Floodwater	High
Fire / severe collapse	Very High

These values represent **relative planning costs**, not guaranteed measurements of real-world danger.

The resulting risk information is used by the route-planning stage.

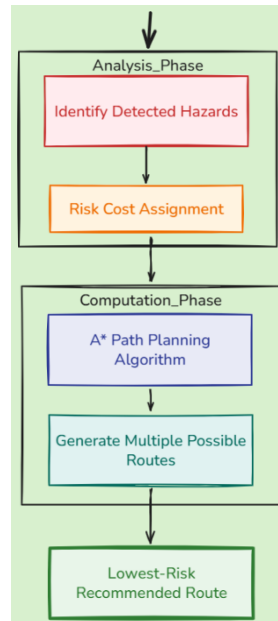
5.1.12. Stage 11 – Risk-Aware Route Planning Using A*

GARUDA uses the **A*** path-planning algorithm to generate a recommended access route based on the mapped environment and assigned risk costs.

The important point is that the system does not simply search for the shortest route.

Instead, the route-planning process considers the risk associated with different areas.

The process is:



The system can therefore provide a **risk-aware recommended access route** to support rescue teams.

This route should be understood as a recommendation based on the information detected by the drone. It is not treated as a guarantee that the route is completely safe.

5.1.13. Stage 12 – Sending Information to the Command Center

The important information generated during the mission is transmitted to the command center when communication is available.

The information may include:

- Drone position
- Survivor detections
- Survivor coordinates
- Hazard locations
- Disaster map
- Risk information
- Recommended access route
- Mission status

The command center provides the rescue team with a centralized view of the information collected by the drone.

This allows human responders to use the aerial information when deciding where and how to conduct the rescue operation.

5.1.14. Stage 13 – Continue Survey or Complete Mission

The system follows a continuous survey process.

If the required area has not yet been covered, the drone continues:

Survey → Capture → Process → Detect → Map → Alert

This process repeats until the planned survey is completed.

Once the mission is complete, the drone can return according to the defined mission procedure.

5.1.15. Complete GARUDA Methodology

The complete methodology can therefore be summarized into the following sequence:

Step 1 – Start Mission

The search-and-rescue mission is initiated for the selected disaster-affected area.

Step 2 – Initialize

The drone takes off and initializes GPS, IMU and other required systems.

Step 3 – Survey

The drone autonomously surveys the target region using a grid/waypoint-based flight pattern.

Step 4 – Sense

RGB and thermal cameras continuously collect information from the environment.

Step 5 – Process

The onboard AI system processes the collected information locally.

Step 6 – Detect

The AI system identifies survivors and defined disaster hazards.

Step 7 – Decision

The system checks whether a survivor or hazard has been detected.

Step 8 – Geo-Tag

Detected survivors and hazards are associated with their geographical locations.

Step 9 – Map

The detections are added to the disaster map.

Step 10 – Assess Risk

Detected hazards are assigned appropriate risk costs.

Step 11 – Plan Route

A* path planning is used to generate a risk-aware recommended access route.

Step 12 – Alert

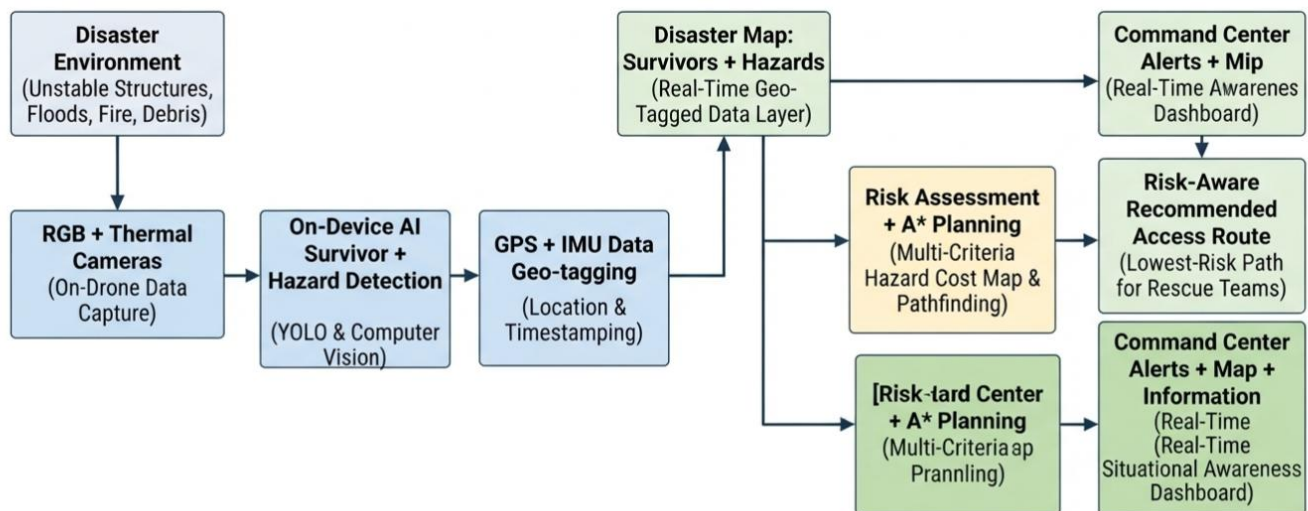
Important information is sent to the command center.

Step 13 – Continue / Complete

If the survey area remains, the drone continues surveying. Otherwise, the mission is completed and the drone returns according to the mission procedure.

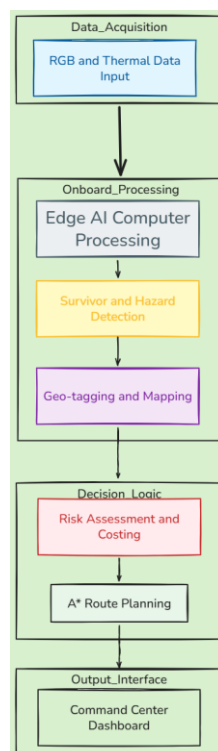
5.1.16.Data Flow in GARUDA

The complete data flow of the system can be represented as:



5.1.17.Role of On-Device Processing in the Workflow

A key characteristic of the GARUDA methodology is that important processing is performed **locally on the drone**.



This approach reduces the dependency on continuous internet connectivity.

If network connectivity is weak or unavailable, the drone can still perform the critical processing locally. Important information can be stored locally and communicated when a suitable connection is available. This is particularly relevant for disaster environments where communication infrastructure may be affected.

5.1.18. Human Role in the GARUDA Workflow

GARUDA is designed to **support rescue teams rather than replace human decision-making**.

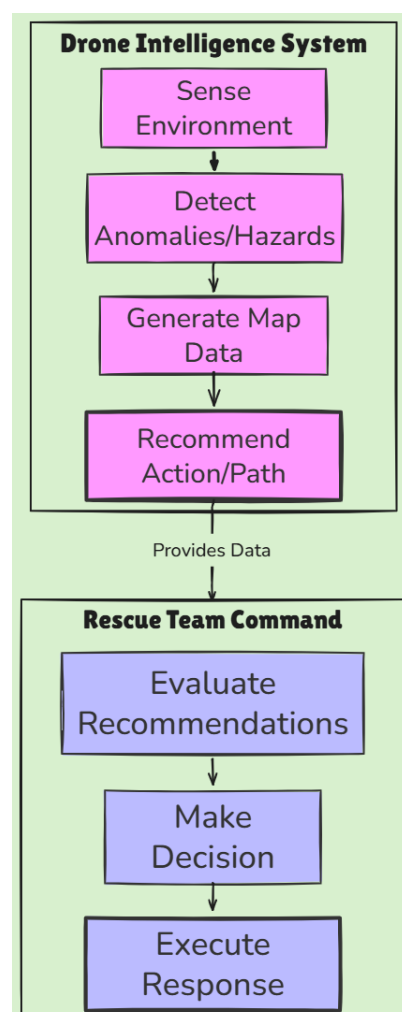
The drone performs the repetitive and potentially dangerous task of surveying the affected area and identifying important information.

The rescue team receives:

- Survivor locations
- Hazard locations
- Disaster map
- Risk information
- Recommended access route

The final rescue decision remains with the responsible human operators.

Therefore, the methodology creates a clear division:



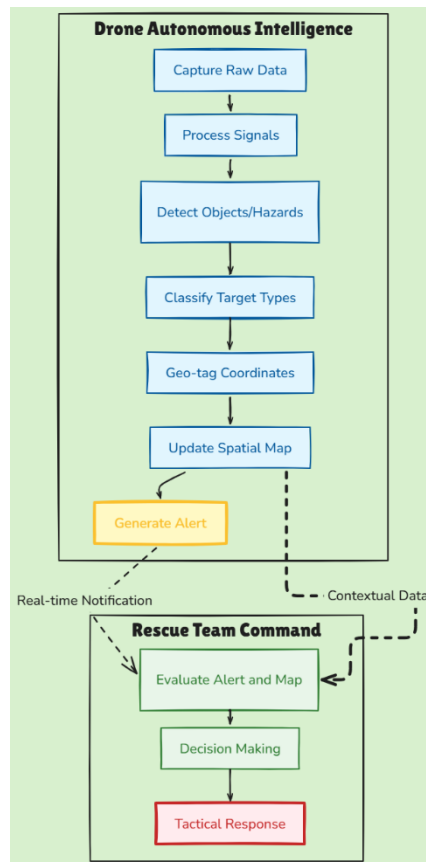
SURVIVOR AND HAZARD DETECTION

Survivor and hazard detection is one of the core functions of **GARUDA**. The purpose of this module is to analyze the information collected by the drone's **RGB and thermal cameras** and identify people as well as important hazards present in a disaster-affected area.

During disasters such as floods, landslides, fires, structural collapses, and debris-related incidents, survivors may be difficult to locate from the ground. At the same time, hazards in the surrounding environment can create additional risks for rescue teams. GARUDA addresses both problems by using onboard AI to analyze the captured sensor data and provide information about detected survivors and hazards.

The detection process takes place locally on the proposed **onboard/edge AI computing system**, allowing the drone to analyze important information without depending entirely on an external cloud server.

The overall detection process is:



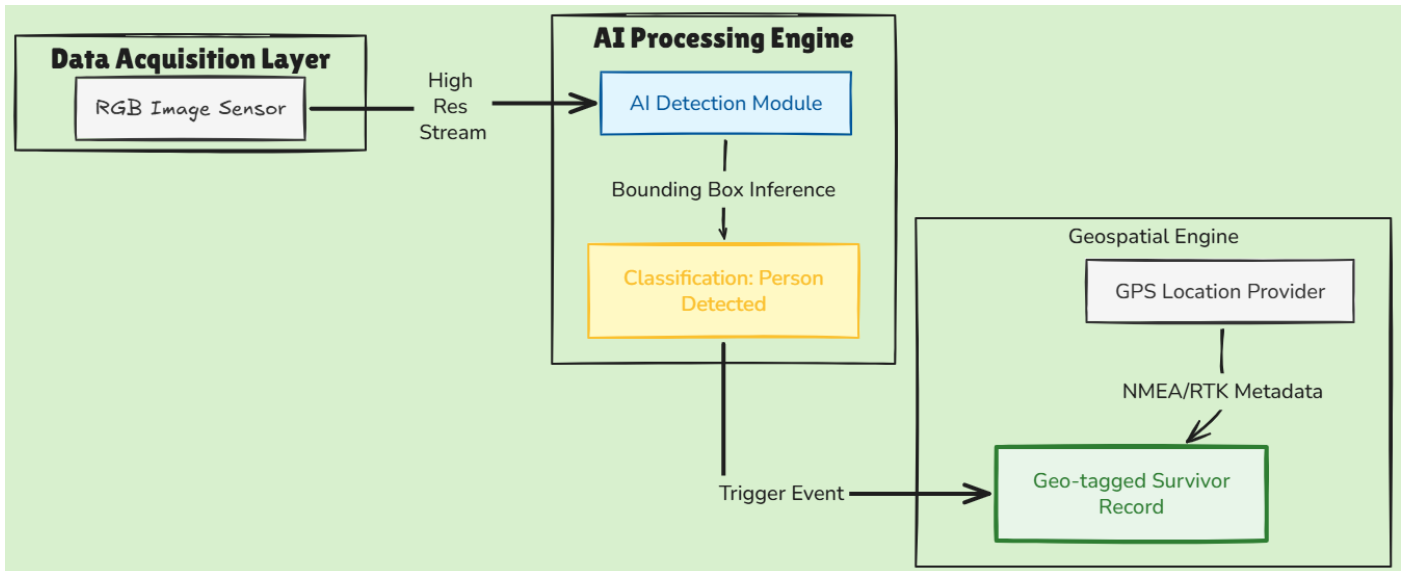
6.1. Survivor Detection

Survivor detection is the primary objective of the GARUDA perception system. The drone continuously captures images while surveying the disaster-affected area, and the onboard AI analyzes these images to identify possible human presence.

The **RGB camera** provides visual information that can be used to identify people based on their appearance, shape, and surrounding visual features.

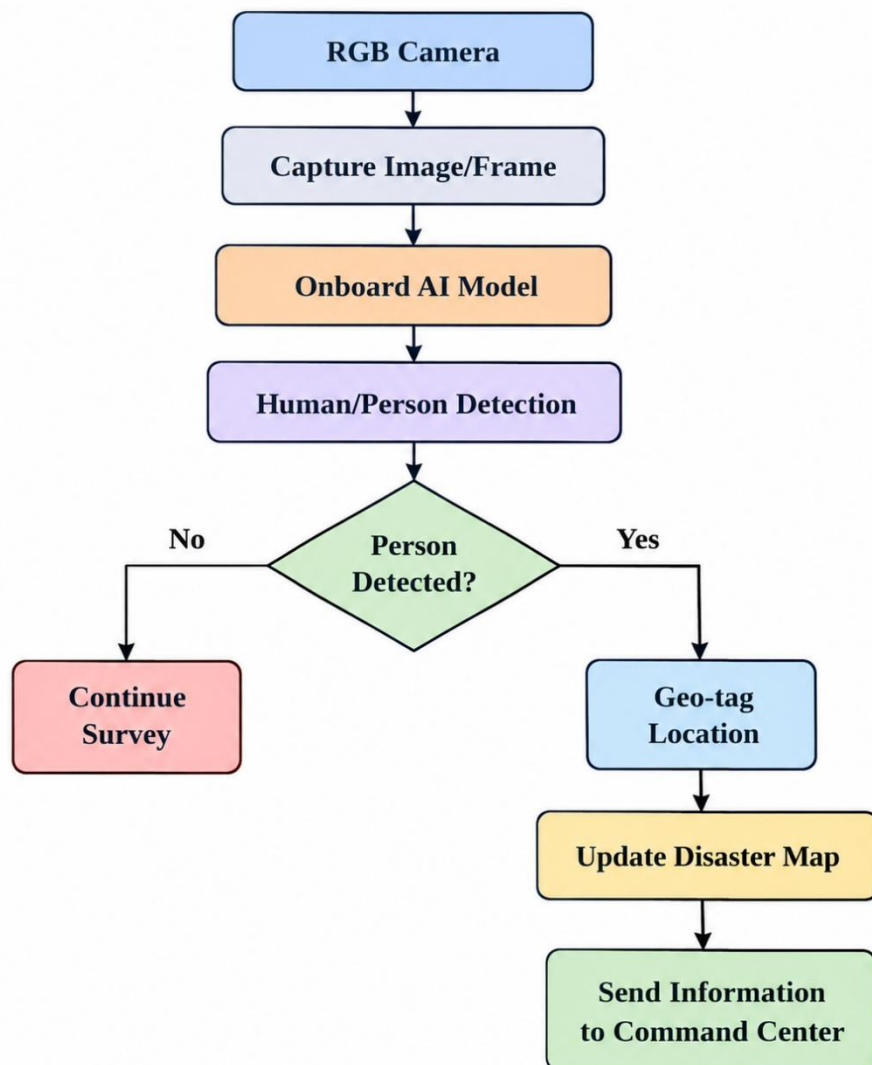
When a person is detected, the system records the detection and associates it with the drone's location information.

The basic process is:



For example, while surveying an affected area, the drone may detect a person on a rooftop, near debris, or in an open area. Instead of only displaying the detection in the camera feed, GARUDA associates the detection with its geographical position. This makes the information more useful for the rescue team.

Survivor Detection Workflow



The system is intended to continuously perform this process during the autonomous survey.

6.2. Thermal-Based Detection

The thermal camera provides an additional source of information for survivor detection.

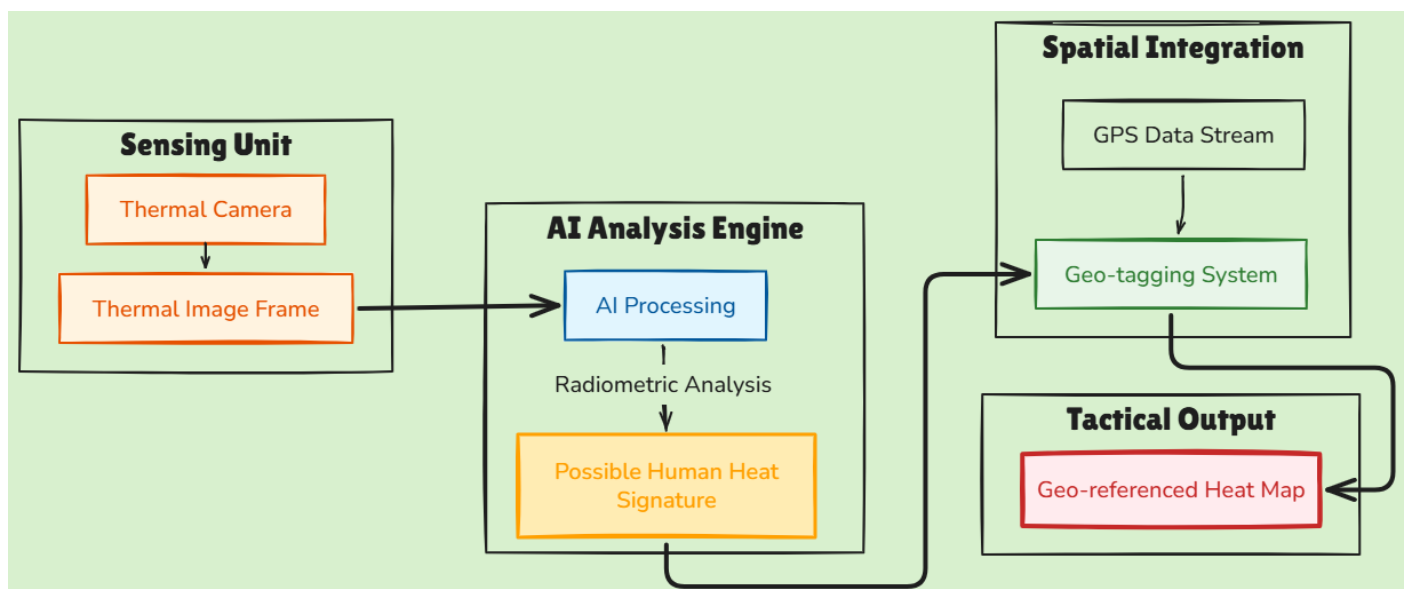
Unlike an RGB camera, which depends on visible-light information, a thermal camera captures information related to **heat signatures**. This makes thermal sensing useful when normal visual detection becomes difficult.

For example, visibility may be affected by:

- Darkness
- Smoke
- Debris
- Poor lighting
- Partial obstruction

In such situations, thermal information can provide additional evidence of possible human presence.

The thermal detection process can be represented as:



The thermal camera is not intended to replace the RGB camera. Instead, both sensors provide complementary information.

For example, an RGB image may provide information about the person's visible surroundings, while thermal information can provide an indication of a heat signature.

This makes the combination of the two sensing methods more useful for disaster search operations than relying on only one sensing modality.

6.3 Hazard Detection

In addition to locating survivors, GARUDA is designed to identify hazards present in the disaster-affected environment.

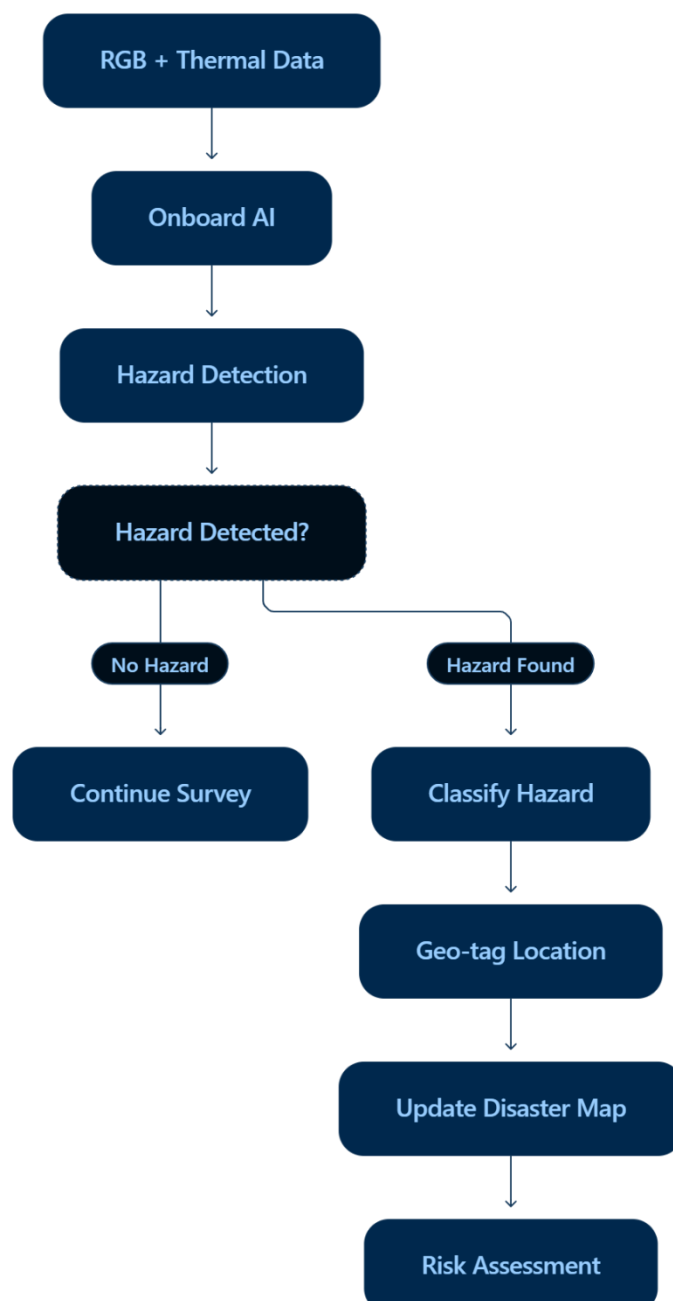
Hazard detection is important because rescue teams need information about dangerous areas before entering them.

The hazards included in the GARUDA project are:

- **Fire**
- **Floodwaters**
- **Damaged structures**
- **Exposed electrical lines**
- **Debris**
- **Landslides**
- **Chemical leaks**

The RGB and thermal sensing systems provide information that can be processed by the onboard AI to identify relevant hazards.

Hazard Detection Workflow



Once a hazard is detected, its location is recorded and added to the disaster map.

The hazard information is then used in the next stages of GARUDA, particularly **risk assessment and risk-aware route planning**.

For example, an area containing significant debris or floodwater can be assigned a higher risk cost than a relatively clear area. This information can later be considered by the A* route-planning process when generating a recommended access route.

6.4. Multi-Modal Sensor Fusion

6.4.1. Introduction

GARUDA uses more than one type of sensor because disaster environments can be unpredictable. A single sensing method may not provide enough information in every situation.

The primary sensing modalities used for detection are:

- **RGB camera**
- **Thermal camera**
- **GPS**
- **IMU**

Each sensor provides different information.

The RGB camera provides visual information, the thermal camera provides heat-signature information, while GPS and IMU provide positional and motion information.

The purpose of **multi-modal sensor fusion** is to combine these different sources of information so that the system can make better-informed detections and associate them with their correct locations.

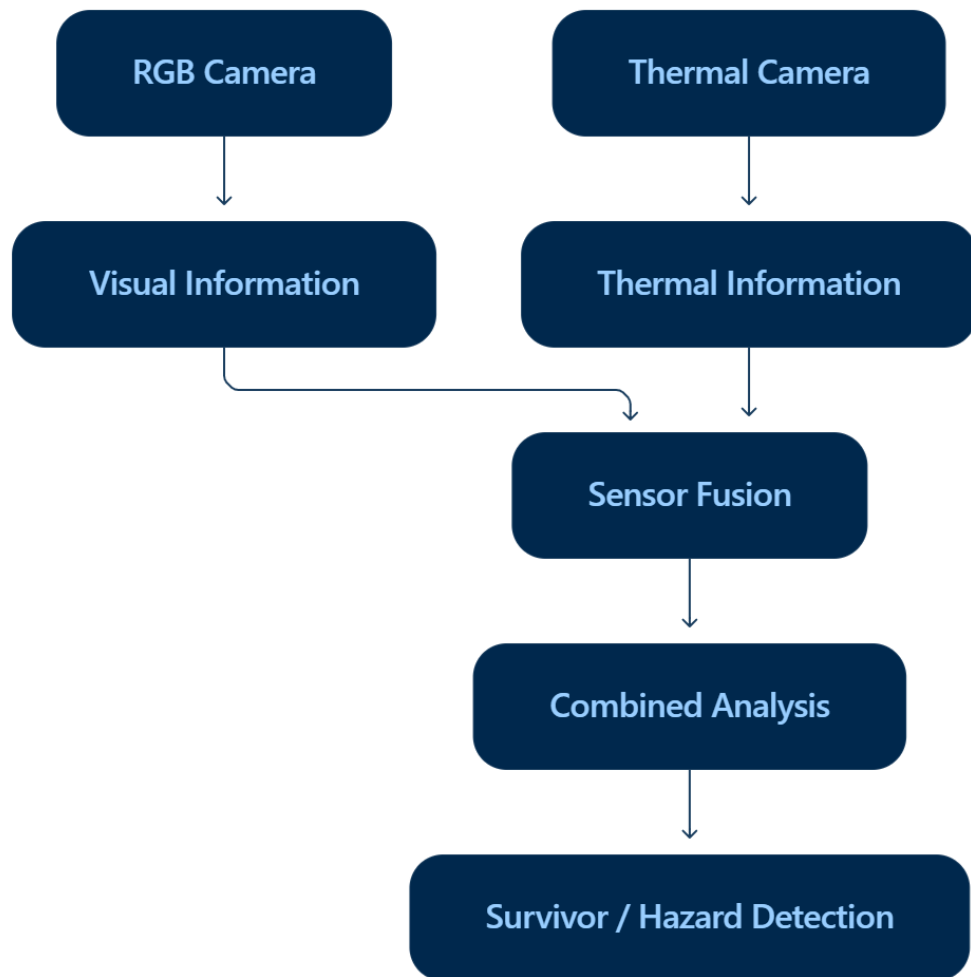
6.4.2. Role of Each Sensor

Sensor	Information Provided	Role in GARUDA
RGB Camera	Visible image information	Survivor and visual hazard detection
Thermal Camera	Heat-signature information	Additional survivor/human-presence information
GPS	Geographic position	Geo-tagging detections
IMU	Motion and orientation	Supports positioning and navigation

The sensors therefore do not perform the same task. Instead, they complement one another.

6.4.3. RGB and Thermal Fusion

The most important sensing combination for survivor detection is **RGB + thermal**.



The RGB camera can provide information about:

- Person appearance
- Surrounding environment
- Buildings
- Roads
- Debris
- Fire and other visible conditions

The thermal camera can provide information about:

- Heat signatures
- Possible human presence
- Thermal differences in the environment

The two data sources can be considered together by the onboard AI system.

For example:

This does not mean that every RGB and thermal frame must always be combined pixel-by-pixel. The exact fusion technique can depend on the final AI implementation.

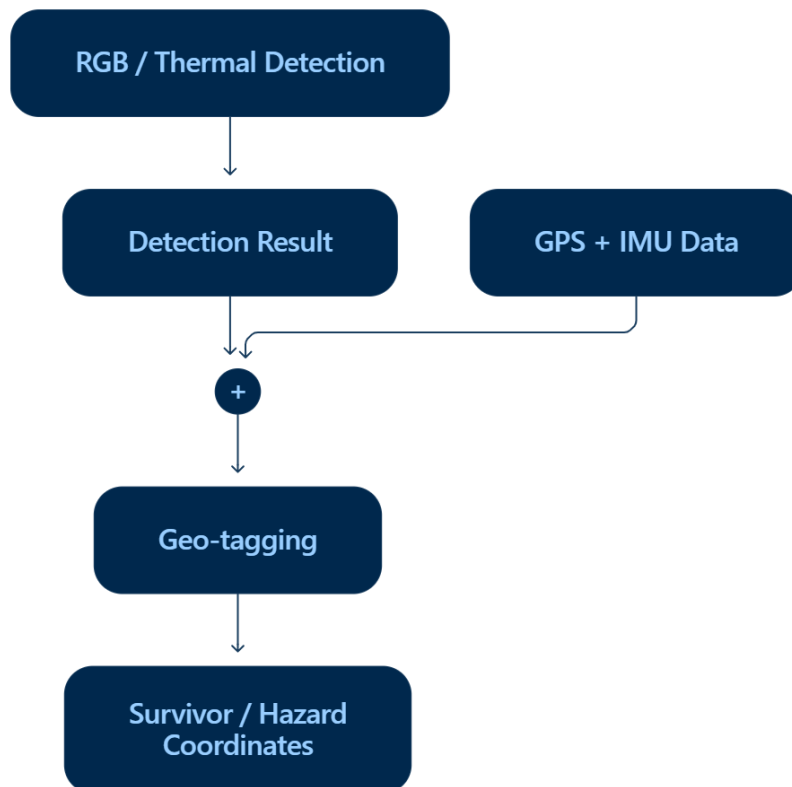
The important concept in GARUDA is that **both sensing modalities contribute to the detection process.**

6.4.4. Location Fusion

Detection alone is not sufficient for a rescue operation. The system also needs to know where the detected survivor or hazard is located.

Therefore, detection information is associated with the drone's positioning information.

The process can be represented as:



For example, if the AI detects a survivor while the drone is at a particular GPS location, that detection can be associated with the corresponding geographical coordinates.

The same approach can be used for detected hazards.

6.4.5. Sensor Fusion for Disaster Mapping

After detections are geo-tagged, the information is added to the disaster map.

The overall process becomes:



The resulting map can show:

Detected survivors, Detected hazards, Drone position, Surveyed region, Risk-related information

As the drone continues surveying, additional detections can be added to the map.

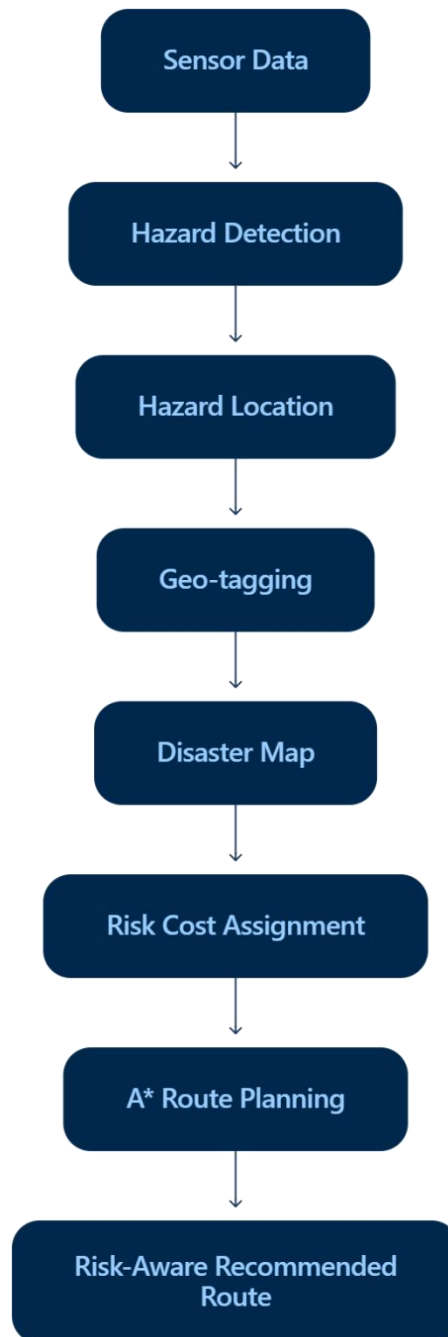
This allows the map to gradually become a representation of the information collected during the mission.

6.4.6. Sensor Fusion for Risk Assessment

The information generated through sensor fusion is also used in GARUDA's risk-assessment process.

Once hazards have been detected and mapped, different risk costs can be assigned to the corresponding areas.

For example:

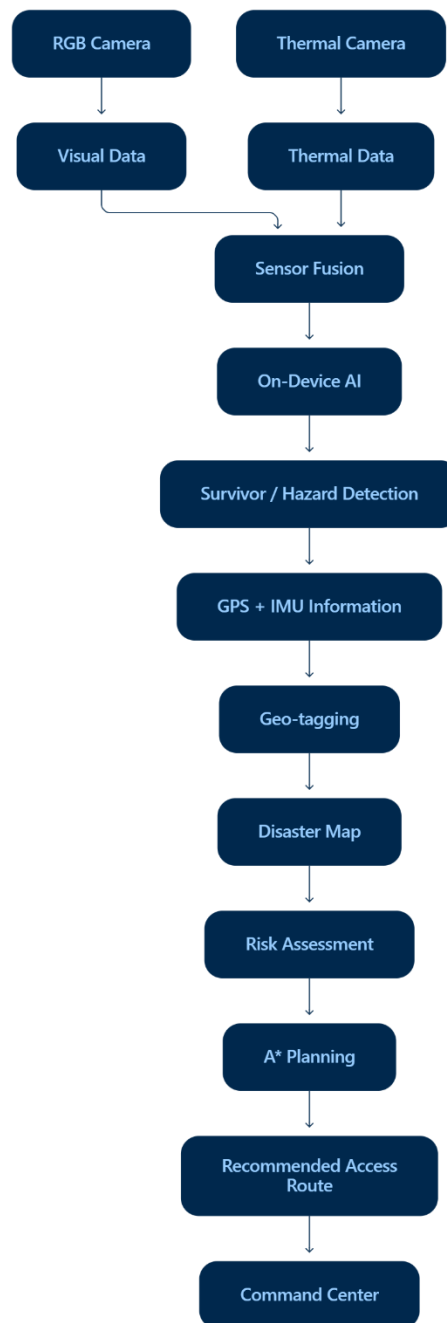


This creates a direct connection between the perception system and the route-planning system.

The drone does not simply detect hazards as isolated objects. The detected information is converted into **location-based information** that can subsequently support route planning.

6.4.7. Complete Detection and Sensor-Fusion Pipeline

The complete GARUDA detection pipeline can therefore be summarized as:



6.4.8. Importance of Multi-Modal Detection in GARUDA

Multi-modal sensing is important because disaster environments are highly variable. Conditions that make visual detection difficult may not necessarily affect thermal information in the same way.

By combining **RGB, thermal, GPS and IMU information**, GARUDA aims to create a more useful representation of the disaster environment.

The key advantage is not simply having more sensors, but **using the information from different sensors together for a common rescue objective**:



This makes multi-modal sensor fusion an important link between GARUDA's **sensing system, AI detection, mapping, risk assessment and command-center workflow**.

AUTONOMOUS NAVIGATION AND MAPPING

Autonomous navigation and mapping are important parts of GARUDA because the drone needs to survey disaster-affected areas without requiring continuous manual control. The navigation system uses **GPS, IMU, and SLAM-based localization** to support movement and position estimation. The information collected during the survey is then used to create a **geo-tagged disaster map** containing detected survivors and hazards.

The navigation and mapping process can be summarized as:



7.1. GPS-Based Navigation

GPS provides GARUDA with the geographical position of the drone during its mission. This information is used to support autonomous movement through the selected disaster-affected area.

After take-off, the drone initializes its GPS and IMU and obtains its initial position. It can then follow a predefined **grid or waypoint-based survey pattern** to cover the target area systematically.

GPS information is also important for connecting detections with their real-world locations. When the AI system detects a survivor or hazard, the drone's position can be used to geo-tag that detection.

The basic process is:



GPS-based navigation therefore supports both **autonomous surveying and location-based disaster information**.

7.2. GPS-Denied Navigation

GPS may not always provide reliable positioning in certain disaster environments. Areas containing damaged structures or other obstructions may make GPS-based navigation difficult.

To address this limitation, GARUDA considers **GPS-denied navigation** using onboard sensor information and visual localization techniques.

When reliable GPS information is unavailable, the system can use information from the **IMU and camera-based perception** to estimate the drone's movement and position.

The purpose is to allow the drone to continue its survey for situations where GPS information is temporarily limited.

The overall concept is:

GARUDA uses **GPS-based navigation** when reliable GPS information is available. The GPS provides the drone's geographical position, which helps it navigate autonomously during the planned survey of the disaster-affected area. However, if GPS signals become limited or unavailable, GARUDA uses **IMU and visual information** to support navigation. In such situations, **SLAM-based localization** can be used to estimate the

drone's position and movement from the surrounding environment. This allows the drone to continue its navigation and survey even when reliable GPS information is not available.

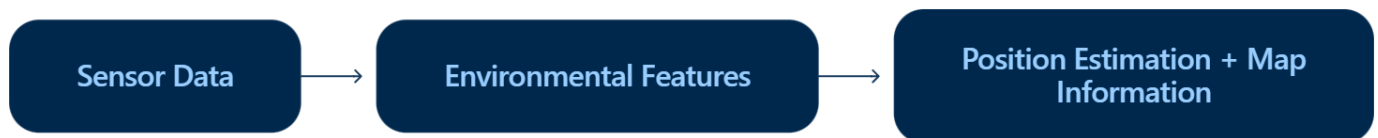
This provides GARUDA with an alternative localization approach instead of making the complete navigation system dependent on GPS.

7.3. SLAM and Localization

SLAM (Simultaneous Localization and Mapping) is considered for situations where the drone needs to estimate its position while building a representation of the surrounding environment.

In GARUDA, SLAM works together with the drone's onboard sensing system and IMU information.

The basic concept is:



As the drone moves through the environment, visual information can be used to identify features in the surroundings. The system uses these observations along with motion information to estimate the drone's movement and position.

SLAM is particularly relevant to GARUDA's GPS-denied navigation requirement because it provides an alternative source of localization when GPS information is not sufficient.

The resulting localization information can also support the mapping process and help associate detected survivors and hazards with the surveyed area.

7.4. Disaster-Area Mapping

Disaster-area mapping is the process of converting the information collected during the drone's survey into a useful representation of the affected region.

GARUDA continuously collects information using its **RGB and thermal cameras**, while GPS and IMU provide positioning and motion information.

When the onboard AI detects a survivor or hazard, the detection is associated with its location through geo-tagging.

The overall process is:



The disaster map can contain:

- Survivor locations
- Hazard locations
- Surveyed areas
- Drone position

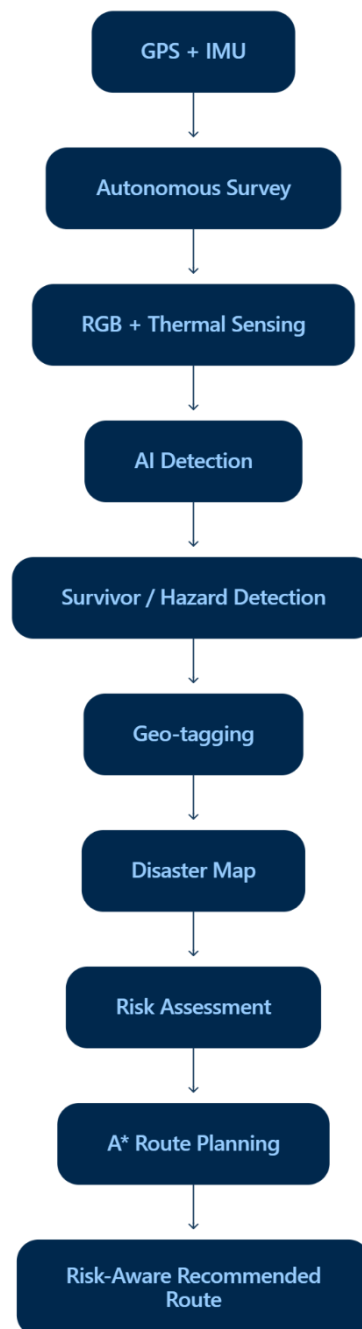
- Risk-related information
- Recommended access-route information

As the drone continues surveying, newly detected information can be added to the map. This provides the command center with an updated view of the disaster-affected area.

The mapped hazard information is also used in the next stage of GARUDA's workflow, where **risk costs are assigned to different areas and A* path planning is used to generate a risk-aware recommended access route.**

Therefore, mapping is not an isolated feature of GARUDA. It connects the drone's **navigation, AI detection, geo-tagging, risk assessment, route planning, and command-center systems.**

Overall Navigation and Mapping Flow



This approach allows GARUDA to move beyond simple aerial observation and provide **location-based disaster intelligence** that can support rescue teams during their response.

RISK ASSESSMENT AND ROUTE PLANNING

Risk assessment and route planning are important parts of GARUDA because detecting a hazard is not enough by itself. The system should also represent **where the hazard is located, how it affects the surrounding area, and which access route can be recommended to the rescue team.**

GARUDA uses the detected hazard information from its RGB and thermal sensing system to update the disaster map. Risk costs are then assigned to different areas of the map, and **A*** path planning is used to generate a **risk-aware recommended access route.**

The overall process is:



8.1. Hazard-Based Risk Mapping

Hazard-based risk mapping is the process of representing detected hazards on the disaster map according to their location and associated risk.

During the drone's autonomous survey, GARUDA uses its **RGB and thermal sensors** to identify relevant hazards such as:

- Fire
- Floodwaters
- Damaged structures
- Exposed electrical lines
- Debris
- Landslides
- Chemical leaks

When a hazard is detected, its location is determined using the drone's positioning information and the detection is geo-tagged.

The process can be represented as:



The mapped hazards are then used to identify areas that may be unsuitable or more difficult for rescue-team access.

For example, an area containing significant debris or floodwater can be represented as a higher-risk region than a relatively clear area.

The purpose of this mapping is to provide the route-planning system with information about the condition of different parts of the surveyed area.

8.2. Risk Cost Assignment

After the hazards are mapped, GARUDA assigns a **risk cost** to different areas of the map.

The risk cost represents the relative difficulty or risk associated with moving through a particular area. Areas with fewer detected hazards can have lower costs, while areas affected by significant hazards can have higher costs.

An example representation is:

Area Condition	Relative Risk Cost
Clear / Normal Area	Low
Debris	Medium
Floodwater	High
Fire / Severe Collapse	Very High

These values are used for **route planning**, rather than being treated as exact measurements of real-world danger.

The purpose of assigning costs is to ensure that the route-planning algorithm considers the detected hazards instead of looking only for the shortest geographical path.

The process is:



This weighted map becomes the input for the A* path-planning stage.

8.3. A* Path Planning

GARUDA uses the **A*** algorithm to generate a route through the mapped environment.

A* is used to search for a suitable path between a starting point and a destination while considering the cost associated with different areas.

In GARUDA, the route-planning process uses the risk costs assigned to the disaster map. The basic workflow is:



Instead of selecting a route only because it is geographically shortest, the system considers the risk costs associated with the areas through which the route passes.

For example, if a shorter route passes through a high-risk flood or debris area while another route is slightly longer but passes through lower-risk areas, the risk-aware planning approach can favor the lower-cost route.

The A* planning process therefore connects the **hazard-detection system** with the **rescue-access planning system**.

8.4. Recommended Access Route

After the A* path-planning process, GARUDA generates a **risk-aware recommended access route** for the rescue team.

The route is based on:

- The surveyed disaster area
- Detected hazards
- Geo-tagged hazard locations
- Assigned risk costs
- A* path planning

The resulting route is displayed as part of the disaster information provided to the command center.

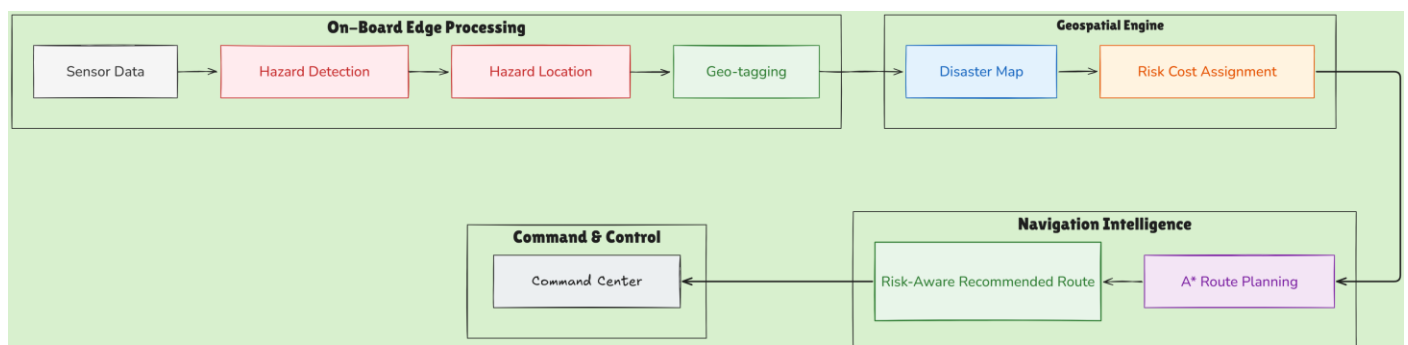
The complete process is:



The purpose of the route is to help rescue teams identify an access path while considering the hazards detected by the drone.

The route should be understood as a **recommendation based on the information available to GARUDA at the time of the survey**, and not as a guarantee that the route is completely safe. Conditions in a disaster area can change after the drone's observation.

Overall Risk-Aware Route Planning Workflow



This module allows GARUDA to move beyond simply detecting and reporting hazards. The detected information is converted into **mapped risk information** and then used to generate a **recommended access route** that can support rescue-team planning.

COMMAND CENTER AND ALERT SYSTEM

The **Command Center and Alert System** is the interface through which rescue teams receive and understand the information collected by GARUDA. The drone performs the survey, detects survivors and hazards, geo-tags the detected locations, and generates disaster-related information. This information is then presented to the command center so that rescue personnel can monitor the situation and plan their response.

The command center does not replace the rescue team. It provides them with **organized, location-based information** collected by the drone to support faster and safer decision-making.

The overall flow is:



9.1. Command-Center Dashboard

The command-center dashboard provides a centralized view of the information generated by GARUDA during the mission.

The dashboard is intended to display the important information received from the drone in an easy-to-understand format.

The main information displayed can include:

- Drone location
- Mission status
- Survivor locations
- Hazard locations
- Disaster-area map
- Risk information
- Recommended access route
- Important alerts

The dashboard receives the processed information from the drone and presents it to the rescue team. Instead of requiring operators to interpret raw camera feeds or sensor data, the system provides the important detected information in a structured form.

The basic dashboard workflow is:



The dashboard therefore acts as the main connection between the autonomous drone system and the human rescue team.

9.2. Survivor and Hazard Visualization

One of the main purposes of the command center is to clearly display the locations of detected survivors and hazards.

When GARUDA detects a survivor, the detection is associated with its geographical location and displayed on the disaster map.

Similarly, when a hazard is detected, its location and relevant information can be represented on the map.

The visualization can therefore contain:

- **Survivor markers** – showing locations where possible survivors have been detected.
- **Hazard markers** – showing locations of detected hazards.
- **Drone position** – showing the current location of GARUDA.
- **Surveyed region** – showing the area covered during the mission.
- **Risk information** – representing areas affected by detected hazards.
- **Recommended access route** – displaying the route generated using the mapped risk information and A* path planning.

The information flow is:



This allows the rescue team to understand the situation spatially instead of receiving isolated detection messages.

9.3. Alerts and Information

GARUDA provides important information to the command center when a survivor or hazard is detected.

The purpose of the alert system is to bring attention to detections that may require action from the rescue team.

An alert can contain relevant information such as:

- Type of detection
- Location/coordinates
- Detection status
- Associated hazard information
- Relevant map information

For example, when a possible survivor is detected, the system can provide the survivor's location to the command center. Similarly, when a hazard such as fire, floodwater, debris, or a damaged structure is detected, its location can be communicated to the rescue team.

The alert workflow is:



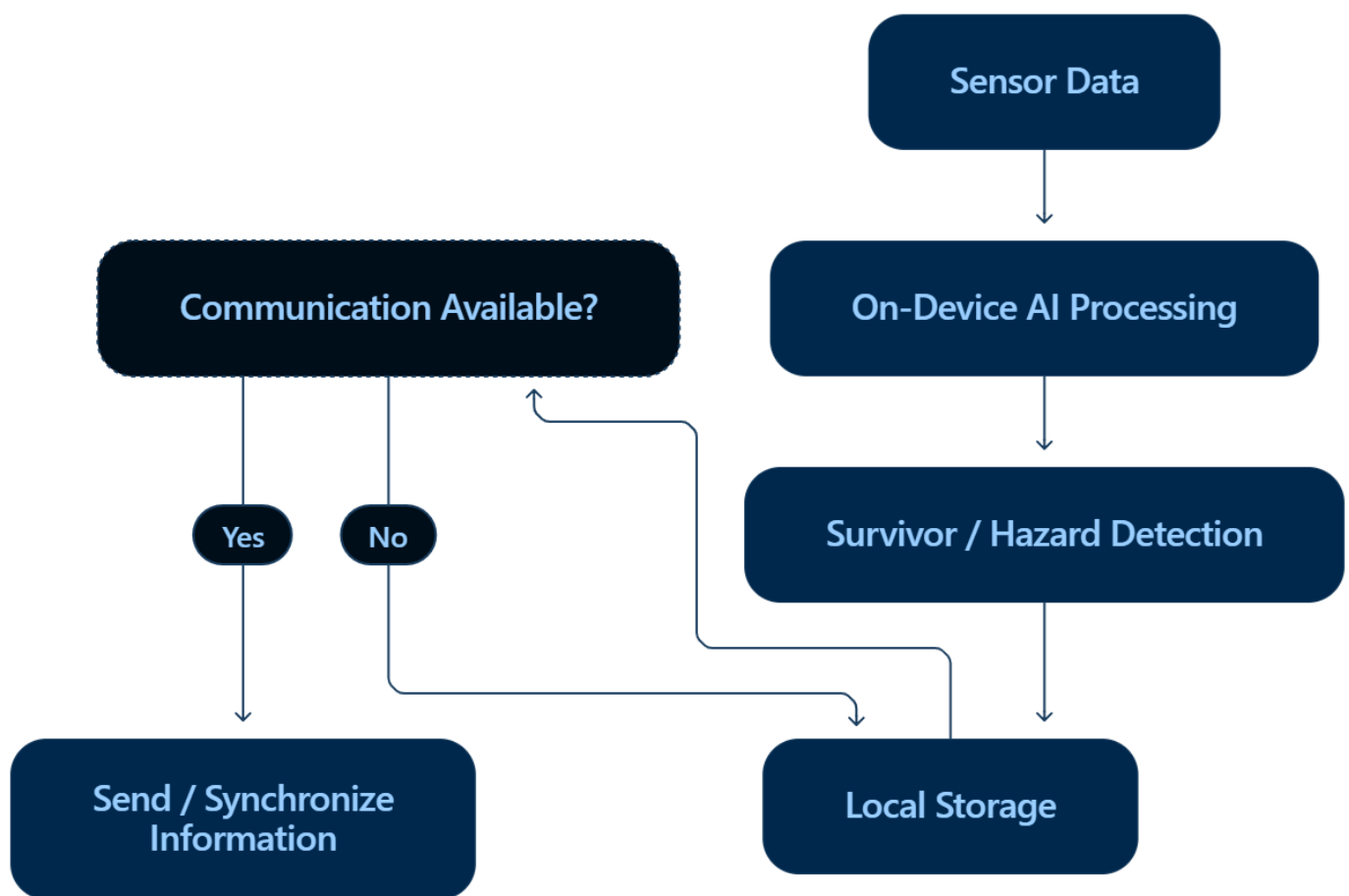
The information generated by GARUDA is intended to **support human decision-making**. The rescue team evaluates the information and decides the appropriate response.

9.4. Offline Operation and Synchronization

A major requirement of GARUDA is the ability to operate when network connectivity is weak or unavailable.

Disaster situations can affect communication infrastructure, making continuous communication with a remote server unreliable. For this reason, GARUDA performs important detection and processing **locally on the drone using the onboard AI system**.

The basic approach is:



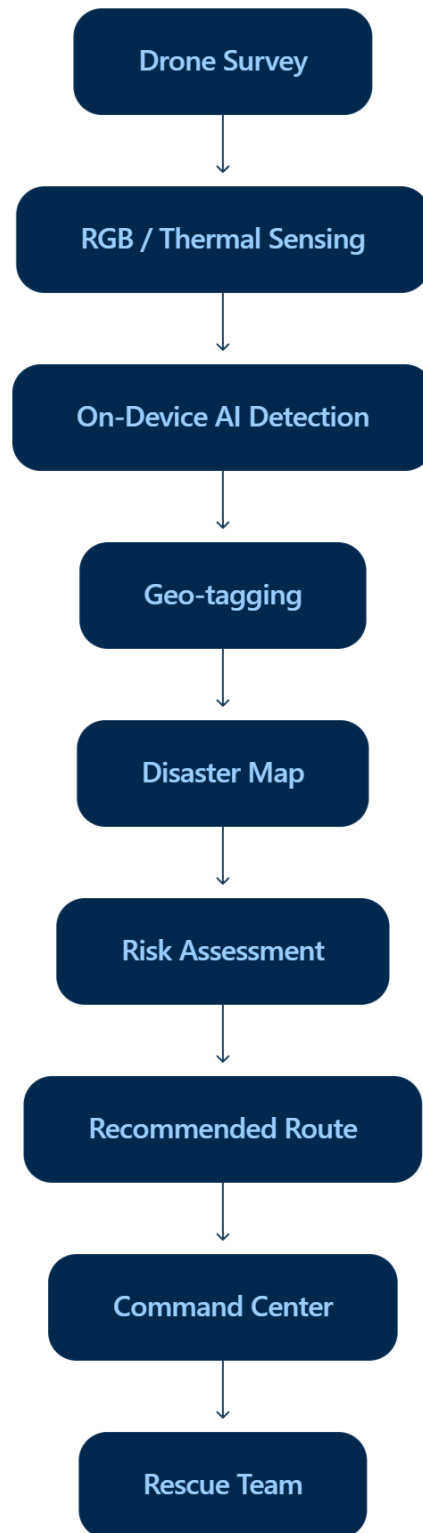
When communication is available, important information such as detections, coordinates, map information and alerts can be transmitted to the command center.

When communication is temporarily unavailable, the critical information can remain stored locally on the drone. Once communication becomes available again, the stored information can be synchronized with the command center.

This provides GARUDA with greater resilience to temporary network limitations without making continuous internet connectivity a requirement for the core detection process.

9.5. Overall Command-Center Workflow

The complete command-center process in GARUDA can be summarized as:



The command center therefore brings together the major outputs of GARUDA—**survivor detections, hazard locations, disaster mapping, risk information, and recommended access routes**—into a single interface for rescue personnel.

The overall purpose is to provide rescue teams with **clear, location-based and actionable information before they enter the disaster-affected area**.

FEASIBILITY ANALYSIS

10.1. Technical Feasibility

GARUDA is technically feasible because the proposed system combines established technologies from computer vision, robotics, embedded AI, autonomous navigation, localization, mapping, and path planning. The system integrates RGB and thermal sensing, GPS and IMU-based localization, on-device AI, survivor and hazard detection, geo-tagging, disaster mapping, risk assessment, A* path planning, communication, and command-center visualization. Each component can be developed and tested independently before being integrated into the complete system. The RGB camera provides visual information for identifying survivors and visible hazards, while the thermal camera provides complementary heat-signature information that can support human detection under challenging visual conditions. On-device AI allows important detection and processing tasks to be performed locally on the drone, reducing dependence on continuous cloud connectivity. GPS can be used for positioning and waypoint navigation, while IMU and visual information can support navigation in situations where GPS is limited. The detected information can be geo-tagged and added to a disaster map, which can then be used for risk assessment and A* path planning to generate a risk-aware recommended access route for rescue teams. Therefore, the technical foundation of GARUDA is achievable through progressive integration and validation of its individual subsystems.

10.2. Hardware Feasibility

The hardware requirements of GARUDA can be implemented using a modular approach based on the existing drone platform. The basic drone system consists of the airframe, motors, ESCs, propellers, flight controller, battery, and power-distribution components required for stable flight. Additional hardware such as RGB and thermal cameras can be mounted on the drone to provide multi-modal sensing. GPS and IMU sensors provide positioning and orientation information required for navigation and geo-tagging. An onboard edge-computing unit is required to process AI models, sensor data, mapping, and other computational tasks locally. A Qualcomm RB5-class platform can be considered as a proposed edge-AI platform, subject to final hardware selection and availability. Communication hardware can be used to transmit telemetry, detections, coordinates, maps, and alerts to the command center when connectivity is available. Since additional sensors and computing hardware increase payload and power consumption, hardware selection must consider weight, battery capacity, flight endurance, thermal management, and mounting requirements. Overall, the required hardware can be progressively integrated with the existing drone platform, with careful attention to payload, power, and environmental constraints.

10.3. Software and AI Feasibility

The software architecture of GARUDA can be developed using established tools and frameworks for computer vision, artificial intelligence, robotics, drone communication, mapping, and visualization. OpenCV can support image processing and camera-data handling, while an object-detection model such as YOLO can be used for survivor and hazard detection. RGB and thermal information can be processed together or through complementary detection pipelines depending on the final system architecture. ROS 2 can provide a modular framework for connecting sensors, AI processing, navigation, localization, mapping, and other robotic components. MAVLink-based communication can support interaction between the flight-control system and higher-level computing systems. GPS and IMU data can be used for localization and geo-tagging, while SLAM can be considered for localization and mapping in GPS-limited environments. Detected survivors and hazards can be associated with their locations and displayed on a disaster map. The map can then be converted into a weighted planning environment in which different conditions are assigned different relative planning costs. A* path planning can use these costs to generate a risk-aware recommended route rather

than simply selecting the shortest path. The actual detection accuracy, processing speed, and navigation performance will depend on the selected hardware, software implementation, datasets, and testing results.

10.4. Operational Feasibility

GARUDA is designed to support disaster-response operations by providing aerial information before rescue personnel enter potentially hazardous areas. A typical mission begins with drone initialization and area surveying, followed by RGB and thermal sensing, on-device AI processing, survivor and hazard detection, geo-tagging, disaster mapping, risk assessment, route recommendation, and transmission of important information to the command center. The command center can display survivor locations, hazard locations, drone status, disaster-map information, risk information, and recommended access routes for rescue personnel. GARUDA is intended as a decision-support system, meaning that the final decision to approach an area or follow a particular route remains with trained rescue personnel. The system also considers situations where communication may be unavailable. Critical information can be processed and stored locally on the drone and synchronized with the command center when communication becomes available. Operational validation should be performed progressively, beginning with component testing and ground testing, followed by controlled flight testing and integrated system testing. Appropriate pre-flight checks, communication-loss handling, battery monitoring, configured failsafe mechanisms, controlled operating conditions, and compliance with applicable drone regulations are also necessary for practical operation.

10.5. Overall Feasibility

Overall, GARUDA can be developed through a phased and modular approach by integrating an existing drone platform with RGB and thermal sensing, onboard edge AI, localization, autonomous navigation, disaster mapping, risk assessment, communication, and command-center visualization. The technologies required for the proposed architecture are available and can be progressively integrated and validated through subsystem testing, ground testing, controlled flight trials, and representative scenarios. The major challenges include limited flight time, additional hardware weight, AI detection reliability, network availability, GPS-denied navigation, environmental conditions, and safe operation. These challenges can be addressed through careful hardware selection, power optimization, sensor fusion, local processing, progressive testing, and appropriate safety mechanisms. Final performance in terms of detection accuracy, flight endurance, autonomous navigation, and route recommendations will depend on the actual hardware, software implementation, environmental conditions, and results obtained during testing. Thus, the proposed GARUDA system presents a practical pathway toward an AI-assisted aerial platform for disaster search, situational awareness, and rescue decision support.

CHALLENGES AND RISK MITIGATION

The development of GARUDA involves several technical and operational challenges because the system is intended to combine autonomous flight, multi-modal sensing, on-device AI, localization, mapping, communication, and rescue-oriented decision support on a single drone platform. Disaster environments are also unpredictable, so the system must be designed to handle limitations in sensing, connectivity, power, and environmental conditions. The following challenges and mitigation strategies are considered as part of the proposed system design.

11.1. Technical Challenges

One of the major technical challenges is integrating multiple subsystems into a single airborne platform. RGB and thermal cameras, GPS, IMU, edge-computing hardware, communication modules, and the flight controller must work together while maintaining reliable data flow and synchronization. AI-based survivor and hazard detection can also be affected by changes in lighting, viewing angle, distance, occlusion, smoke, debris, and other visual conditions. Thermal detection may provide complementary information, but its performance can also vary depending on environmental conditions and the characteristics of the target. Autonomous navigation presents another challenge because disaster areas may contain changing obstacles, damaged structures, narrow spaces, and unpredictable conditions. GPS may also become unreliable in certain environments, requiring alternative localization approaches such as visual information, IMU data, or SLAM where technically suitable. Integrating all these functions while maintaining real-time performance on an onboard computing platform is therefore an important engineering challenge.

11.2. Environmental Challenges

Disaster environments can be significantly different from normal drone operating conditions. Heavy rain, strong wind, dust, smoke, high temperatures, poor visibility, and water can affect both the drone and its sensors. Damaged buildings, trees, electrical infrastructure, debris, and other obstacles can also make autonomous navigation more difficult. In addition, environmental conditions can change during a mission, meaning that information collected at one point may become outdated as the situation develops. RGB cameras may experience reduced visibility because of darkness, smoke, or obstructions, while thermal cameras can also be affected by environmental and surface-temperature conditions. These factors make it necessary to validate the system under controlled and progressively more challenging conditions rather than assuming that detection and navigation performance will remain constant in every environment.

11.3. Power and Flight-Time Constraints

Limited battery capacity is an important constraint for any drone-based system. GARUDA carries additional components such as RGB and thermal cameras, an edge-computing unit, communication hardware, and other sensors, which increase the overall payload and power consumption. Continuous AI processing and sensor operation can further increase energy usage. Higher payload can also affect flight performance and reduce available flight time. Therefore, the system must balance computing requirements, sensor operation, payload weight, and battery capacity. Efficient AI models, selective sensor usage, optimized processing, appropriate mission planning, and careful hardware selection can help reduce unnecessary power consumption. The system can also prioritize important detections and information rather than continuously transmitting all raw sensor data.

11.4. Connectivity Challenges

Disaster situations can involve damaged communication infrastructure, weak cellular networks, network congestion, or complete loss of connectivity. A system that depends entirely on cloud-based processing or continuous internet access may therefore become unreliable in such conditions. GARUDA addresses this

challenge by performing critical AI processing locally on the drone. Detection results, coordinates, and other important information can also be stored locally when communication is unavailable. Once a communication link becomes available, the stored information can be synchronized with the command center. This approach allows the drone to continue collecting and processing important information even during temporary communication loss, although real-time remote monitoring will naturally depend on the availability and quality of the communication link.

11.5. Mitigation Strategies

GARUDA follows a phased and modular approach to reduce the risks associated with these challenges. Individual hardware and software components can first be tested independently before being integrated into the complete system. Ground testing can be used to validate RGB and thermal sensing, AI detection, GPS and geo-tagging, local processing, storage, and communication before progressing to controlled aerial testing. Sensor fusion between RGB, thermal, GPS, and IMU data can provide complementary information and improve the overall perception pipeline. AI models can be evaluated using representative datasets and controlled test scenarios before being considered for more challenging environments. Power consumption can be managed through lightweight hardware, optimized AI models, selective sensor usage, and appropriate mission planning. Local processing and storage provide resilience against communication failures, while synchronization can occur when connectivity is restored. Navigation can use GPS when available and alternative localization approaches where appropriate. Pre-flight checks, battery monitoring, communication-loss handling, configured flight-controller failsafes, controlled operating boundaries, and human supervision can further reduce operational risks. Most importantly, the system should be validated progressively through **component testing → ground testing → controlled flight testing → integrated testing → representative scenario testing**, allowing identified limitations to be addressed before advancing to more demanding conditions.

TESTING AND EVALUATION PLAN

Testing and evaluation are essential for validating the functionality, reliability, and integration of the GARUDA system. Since the proposed solution combines drone hardware, RGB and thermal sensing, on-device AI, localization, mapping, route planning, and communication, testing will be performed progressively rather than evaluating the complete system at once. The testing process will begin with individual component and ground-level tests, followed by controlled flight tests and integrated system evaluation. This approach allows problems to be identified and corrected at each stage before moving to more complex testing conditions.

12.1. Detection Testing

Detection testing focuses on evaluating GARUDA's ability to identify survivors and relevant hazards using RGB and thermal sensor data. Initial testing will be performed on the ground using controlled scenes and known targets. The RGB camera will be tested for its ability to capture clear visual information, while the thermal camera will be evaluated for its ability to capture heat signatures. The AI detection pipeline will then be tested using the captured sensor data to determine whether predefined survivor and hazard classes can be detected correctly. Testing will consider different distances, viewing angles, lighting conditions, partial occlusion, and other representative conditions where applicable. The evaluation can use measures such as detection accuracy, precision, recall, false detections, missed detections, and processing time. These results will help determine the suitability of the detection pipeline before integrating it with the airborne system.

12.2. Navigation Testing

Navigation testing evaluates the drone's ability to move according to the planned mission while maintaining stable and controlled flight. Initial tests will be conducted in a controlled environment to verify takeoff, landing, basic movement, waypoint navigation, and position estimation. GPS and IMU data will be monitored during these tests to verify that the navigation system receives appropriate positional and orientation information. Autonomous survey patterns such as waypoint-based or grid-based movement can then be tested over a predefined area. Where GPS-denied navigation is part of the implemented prototype, visual information, IMU data, and SLAM-based localization can be evaluated separately under controlled conditions. The evaluation can consider waypoint accuracy, position deviation, navigation stability, obstacle response where implemented, and successful completion of the planned survey path.

12.3. Mapping and Geo-Tagging Testing

Mapping and geo-tagging testing verifies whether detections obtained by the drone are correctly associated with their geographical locations and represented on the disaster map. During testing, known target locations can be established within a controlled test area. When the AI system detects a survivor or hazard, the corresponding GPS and positional information will be recorded and associated with the detection. The resulting information will then be displayed on the disaster map. The testing will evaluate whether detections appear at the correct locations, whether multiple detections can be represented simultaneously, and whether the map is updated correctly as the drone continues its survey. Where appropriate, the difference between the known target location and the recorded geo-tagged location can be measured to evaluate localization error.

12.4. Route-Planning Testing

Route-planning testing evaluates the ability of GARUDA to generate a risk-aware recommended access route using the information collected during the drone survey. Detected hazards will first be placed on a test map and assigned relative planning costs based on their predefined risk categories. The resulting weighted map will then be provided to the A* path-planning algorithm. Different test scenarios can be created by changing

the location and severity of mapped hazards. The system will be evaluated to determine whether the generated route avoids or accounts for high-cost areas where appropriate and whether it can identify an alternative route when the direct path contains significant risk costs. Testing can compare route distance, accumulated planning cost, and computational time. The resulting route will be treated as a **recommended access route for decision support**, rather than a guarantee that the route is physically safe.

12.5. System Integration Testing

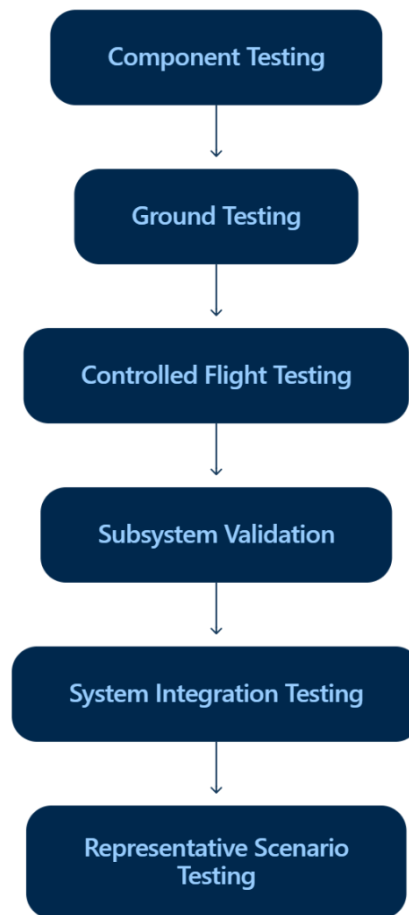
System integration testing evaluates the complete GARUDA workflow after the individual subsystems have been validated. The integrated test will follow the complete processing chain:



The test will verify that sensor data can be captured during a mission, processed locally by the onboard AI system, and converted into survivor or hazard detections. The detected information will then be geo-tagged and transferred to the disaster map, where risk information and route planning can be generated. The resulting information will be displayed through the command-center interface.

System integration testing will also examine communication-loss conditions. When communication is available, important information should be transmitted to the command center. When communication is unavailable, critical information should remain available through local processing and storage and should be synchronized when connectivity is restored. The final evaluation will consider the reliability of the complete data pipeline, response time, synchronization, detection-to-map consistency, navigation behavior, and overall system stability.

Overall Testing Progression



This progressive testing methodology ensures that GARUDA's individual capabilities are validated before they are combined into a complete disaster-response system. Final performance values such as detection accuracy, navigation error, processing latency, and flight endurance should be reported only after the corresponding tests have been conducted and measured.

IMPACT AND BENEFITS

GARUDA is designed to support disaster-response operations by providing aerial intelligence before rescue personnel enter potentially hazardous areas. By combining autonomous surveying, RGB and thermal sensing, on-device AI, geo-tagging, disaster mapping, risk assessment, and command-center visualization, the system aims to provide rescue teams with timely and organized information about survivors, hazards, and affected areas. The expected impact of GARUDA can be considered across social, rescue-team, economic, and environmental aspects.

13.1. Social Impact

GARUDA can contribute to faster identification of survivors in disaster-affected areas by providing aerial surveillance and AI-assisted detection. In situations such as floods, landslides, fires, earthquakes, and structural collapse, some affected locations may be difficult or dangerous for ground teams to access. An aerial platform can survey such areas and provide information about possible survivor locations before responders physically enter the region. The use of both RGB and thermal sensing provides complementary information that can support detection under different visibility conditions. Geo-tagging detected locations and displaying them on a disaster map can also improve situational awareness and help organize the response. By reducing the need for responders to immediately enter unknown or hazardous areas for initial assessment, GARUDA has the potential to support safer and more informed disaster-response operations.

13.2. Rescue-Team Benefits

GARUDA is designed primarily as a decision-support tool for rescue teams. Instead of relying only on information collected through ground inspection, responders can receive aerial information about the affected region. The system can identify potential survivors and hazards, associate them with geographic locations, and display the information through a command-center dashboard. Hazard information can be used to assess the relative risk of different areas and generate a **risk-aware recommended access route** using A* path planning. This can help responders consider mapped hazards when planning access to a location. The system can also continue important processing locally when communication is temporarily unavailable and synchronize stored information when connectivity is restored. These capabilities can improve situational awareness, support prioritization, and reduce unnecessary exposure of rescue personnel to unknown conditions. Final operational decisions, however, remain with trained rescue personnel.

13.3. Economic Benefits

GARUDA can provide economic benefits by building upon an existing drone platform and using a modular architecture rather than requiring a completely new aerial system. Existing flight hardware can be progressively enhanced with sensing, edge-computing, AI, communication, and mapping capabilities according to the requirements of the prototype. A reusable drone platform can potentially support multiple disaster-response missions after appropriate testing and maintenance. On-device processing can also reduce dependence on continuous cloud infrastructure for critical detection tasks. By providing aerial information during the initial assessment phase, the system may help reduce the time and resources required for manual inspection of large or difficult-to-access areas. The economic benefit will ultimately depend on the final hardware configuration, operating costs, maintenance requirements, and actual deployment conditions.

13.4. Environmental Benefits

As a battery-powered aerial platform, GARUDA can conduct aerial assessment without requiring continuous movement of ground vehicles through the affected area. Remote surveying can reduce the need for unnecessary ground movement during the initial assessment of difficult or inaccessible regions. The system can also support disaster assessment in areas affected by floods, landslides, fires, and structural damage

without requiring immediate physical access by responders. Multi-modal sensing and AI-based analysis can help focus attention on relevant locations rather than requiring complete manual inspection of every area. However, the environmental impact of the system also depends on battery production and disposal, electronic components, maintenance, and operational practices. Therefore, responsible battery management, component reuse, and appropriate maintenance should be considered throughout the system's lifecycle.

Overall Benefits

GARUDA provides several important benefits for disaster-response operations by combining RGB and thermal sensing, on-device AI, geo-tagging, disaster mapping, and risk-aware route planning. It can help rescue teams detect potential survivors and hazards more quickly while reducing the need for responders to immediately enter unknown or dangerous areas. Since critical AI processing is performed locally on the drone, the system can continue processing important information even when network connectivity is limited, with data stored locally and synchronized when communication is restored. Geo-tagged detections allow survivors and hazards to be accurately represented on a disaster map, giving rescue teams better situational awareness. The system can also use detected hazards to generate a risk-aware recommended access route, helping responders consider mapped risks while planning their approach. Overall, GARUDA aims to support faster detection, improved responder safety, better situational awareness, more informed rescue decisions, and efficient disaster assessment.

Overall Impact

GARUDA aims to improve disaster-response decision-making by providing faster aerial situational awareness, identifying potential survivors and hazards, mapping their locations, and providing risk-aware information to rescue teams. Its primary impact is to help responders understand the environment before entering it, supporting safer, more informed, and more efficient rescue operations.

SWOT ANALYSIS

SWOT analysis provides a structured view of GARUDA's internal capabilities and limitations, along with external opportunities and potential risks. It helps identify the factors that can influence the development, testing, and future deployment of the proposed AI-powered disaster-response drone.

14.1. Strengths

GARUDA's primary strength is its combination of **autonomous aerial surveying, RGB and thermal sensing, and on-device AI** in a single disaster-response platform. The use of RGB and thermal cameras provides complementary information for survivor and hazard detection, while local AI processing reduces dependence on continuous internet connectivity. The system also combines detection with **geo-tagging and disaster mapping**, allowing detected survivors and hazards to be associated with their locations. Another strength is the proposed **risk-aware route planning**, where mapped hazards can be assigned relative planning costs and processed using A* to generate a recommended access route. The modular architecture also allows the system to be progressively upgraded with additional sensors, computing capabilities, and navigation functions.

14.2. Weaknesses

GARUDA has several technical limitations that must be considered during development. Drone flight time is constrained by battery capacity, particularly when additional cameras, edge-computing hardware, and communication equipment increase the payload and power consumption. AI-based detection can also be affected by environmental conditions, camera quality, occlusion, smoke, darkness, and other factors, meaning that detection results require proper validation. Thermal sensing provides complementary information but may also have limitations depending on environmental and surface-temperature conditions. Autonomous navigation becomes more challenging in complex or changing environments, particularly where GPS is unreliable or unavailable. The system also requires technical expertise for hardware integration, AI development, drone operation, testing, and maintenance.

14.3. Opportunities

GARUDA has opportunities for further development in disaster management and emergency-response applications. The platform can potentially be adapted for different scenarios such as **floods, earthquakes, landslides, fires, structural collapse, and other hazardous environments**. Future versions can incorporate additional sensing technologies, improved AI models, advanced localization, and more capable edge-computing hardware. The system can also be extended toward **multi-drone coordination**, where multiple drones survey different regions and share relevant detections to improve coverage. Integration with larger disaster-management networks, ground sensors, satellite information, and centralized command systems could further expand its capabilities.

14.4. Threats

The operation of GARUDA can be affected by external factors such as **drone regulations, restricted airspace, environmental conditions, communication failures, hardware failures, and rapidly changing disaster situations**. Extreme weather conditions such as strong winds, heavy rain, smoke, dust, or high temperatures may affect flight performance and sensor reliability. Loss of communication can limit real-time transmission to the command center, although local processing and storage are intended to reduce this dependency. Hardware damage or battery failure can interrupt a mission and therefore requires appropriate safety and failsafe mechanisms. AI systems may also produce false detections or miss targets under challenging conditions, making human verification important for critical decisions. Regulatory requirements and operational restrictions may further affect where and how the drone can be tested or deployed.

FUTURE SCOPE

GARUDA is designed as a modular platform, which allows its capabilities to be expanded beyond the initial prototype. Future development can focus on increasing the coverage area, improving perception and navigation, integrating additional sources of disaster information, and enabling better coordination between aerial systems and emergency-response teams. The following areas represent potential directions for future development.

15.1. Multi-Drone Coordination

A future version of GARUDA can be extended from a single-drone system to a **multi-drone disaster-response network**. Multiple drones could operate over different sections of a large disaster-affected region, allowing the survey area to be divided among them. Each drone could independently collect RGB and thermal data, perform onboard detection, and geo-tag identified survivors or hazards. Relevant information could then be shared between drones or with a central command system to create a combined disaster map. Multi-drone coordination could also allow different drones to perform different tasks, such as wide-area surveying, detailed inspection, or focused survivor search. Such a system would require further development of task allocation, communication, collision avoidance, and coordinated mission planning.

15.2. Advanced Sensor Integration

Future versions can incorporate additional sensors to improve GARUDA's ability to understand complex disaster environments. Possible additions include LiDAR or depth sensors for improved obstacle detection and three-dimensional mapping, improved thermal cameras for enhanced heat-signature analysis, environmental sensors for detecting specific conditions, and audio or microphone systems for identifying possible human voices or distress signals. These sensors could be combined with the existing RGB and thermal cameras through multi-sensor processing. Additional sensing should be introduced based on mission requirements while considering payload, power consumption, processing requirements, and flight endurance.

15.3. Improved Autonomous Navigation

The navigation system can be further developed to operate more effectively in dynamic and GPS-limited environments. Future improvements could include more robust visual-inertial localization, improved SLAM, advanced obstacle detection and avoidance, adaptive survey planning, and dynamic replanning when new hazards are detected. Instead of following only a predefined survey path, the drone could adapt its mission based on the information collected during the flight. For example, detection of a potential survivor could trigger a more detailed inspection of the surrounding area, subject to safety and mission constraints. Navigation improvements would require extensive testing because disaster environments can contain unpredictable obstacles and rapidly changing conditions.

15.4. Disaster-Response Ecosystem

In the longer term, GARUDA can evolve from an individual drone platform into a broader **disaster-response ecosystem**. The system could potentially integrate drone data with ground sensors, satellite information, geographic information systems, emergency communication networks, and centralized disaster-management platforms. A unified command interface could combine survivor detections, hazard information, aerial imagery, disaster maps, and other available information to provide a broader operational picture. Integration with multiple response agencies could also support information sharing and coordination, subject to appropriate technical, organizational, privacy, and regulatory requirements. The long-term vision is to create an intelligent aerial information layer that helps emergency teams understand changing disaster conditions and make more informed response decisions.

CONCLUSION

GARUDA presents a proposed AI-powered autonomous drone system designed to support search-and-rescue operations in disaster-affected and hazardous environments. The system combines **RGB and thermal sensing, on-device AI, GPS/IMU-based localization, autonomous surveying, geo-tagging, disaster mapping, risk assessment, and route planning** to provide rescue teams with timely and organized information about survivors and environmental hazards.

A key aspect of GARUDA is its use of **on-device processing**. Instead of depending entirely on cloud connectivity, critical sensor data and AI-based detection can be processed locally on the drone. This approach is intended to support low-latency operation and allow important information to be retained even when network connectivity is weak or temporarily unavailable. Detected survivors and hazards can be geo-tagged and represented on a disaster map, providing a clearer view of the affected area.

The proposed risk-assessment and route-planning module further extends the system beyond basic detection. By assigning relative planning costs to mapped hazards and applying the *A path-planning algorithm**, GARUDA can generate a **risk-aware recommended access route** for rescue teams. This route is intended as decision-support information and does not represent a guarantee of physical safety.

The feasibility analysis indicates that GARUDA can be developed through a **modular and phased approach**. Individual components can first be validated through component and ground testing, followed by controlled flight testing, subsystem validation, system integration, and representative scenario testing. This approach allows technical limitations to be identified and addressed progressively before more complex deployment scenarios are considered.

GARUDA also provides opportunities for future expansion through **multi-drone coordination, advanced sensor integration, improved autonomous navigation, and integration with a wider disaster-response ecosystem**. These developments could extend the system from a single aerial platform into a coordinated network for disaster sensing and information sharing.

Overall, GARUDA aims to shift the initial disaster-response process from **“rescuers enter first and assess the situation”** toward **“the aerial system surveys, detects, maps, and informs before responders enter.”** By providing early information about survivors, hazards, affected regions, and recommended access routes, the proposed system can serve as a technological support layer for safer and more informed disaster-response operations.

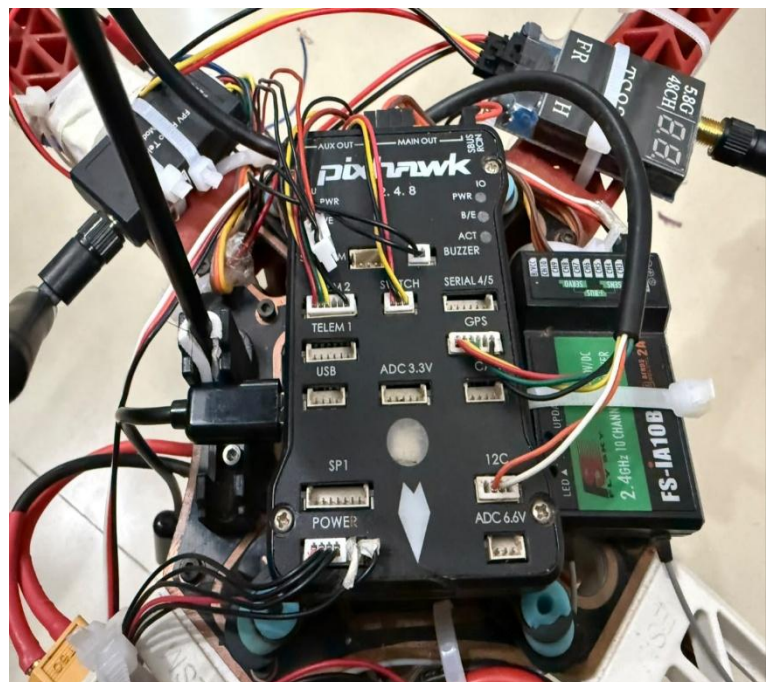
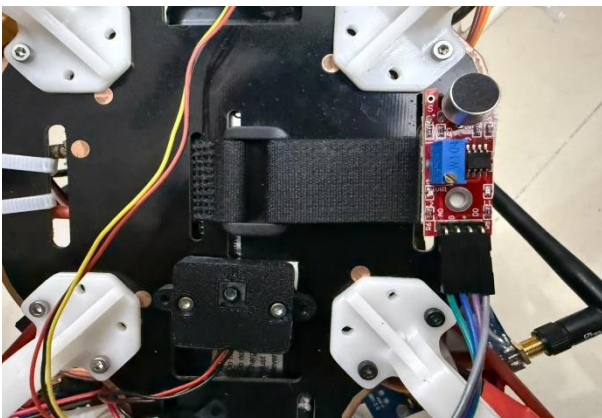
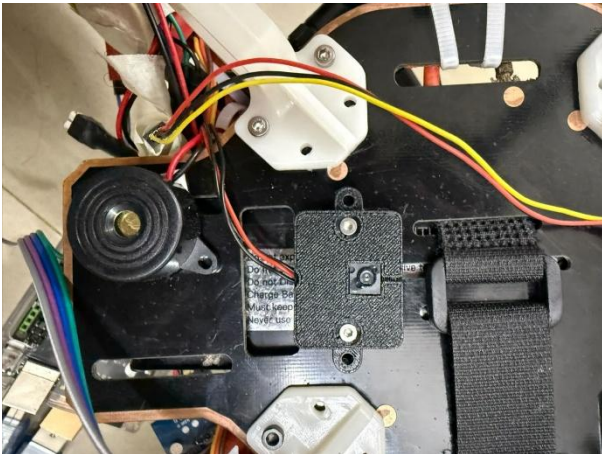
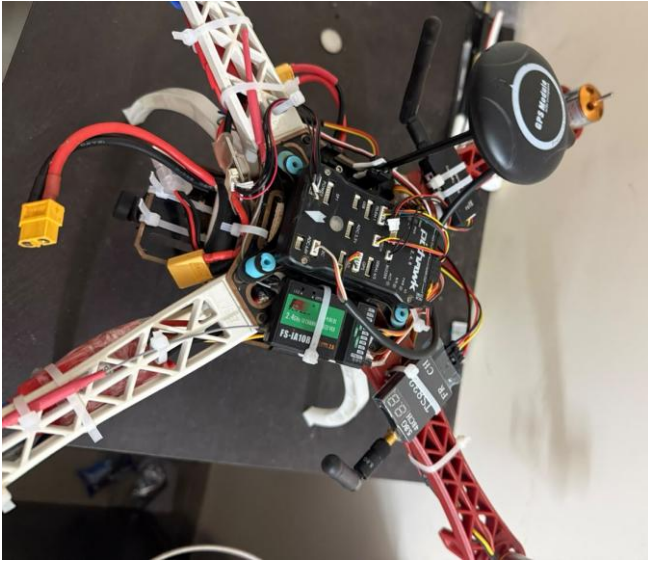
GARUDA — Let the drone enter the danger zone before the rescuer does.

REFERENCES

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ADDITIONAL PICTURES





Mission Planner 1.3.83 build 1.3.9384.38258 ArduCopter V4.5.4 (fd1bcc61) on fmuw3 00200020 35335111 34383632

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Home: 2367.86 m

ARDUPILOT COM9 57600 COM9-T-QUADROTOR CONNECT

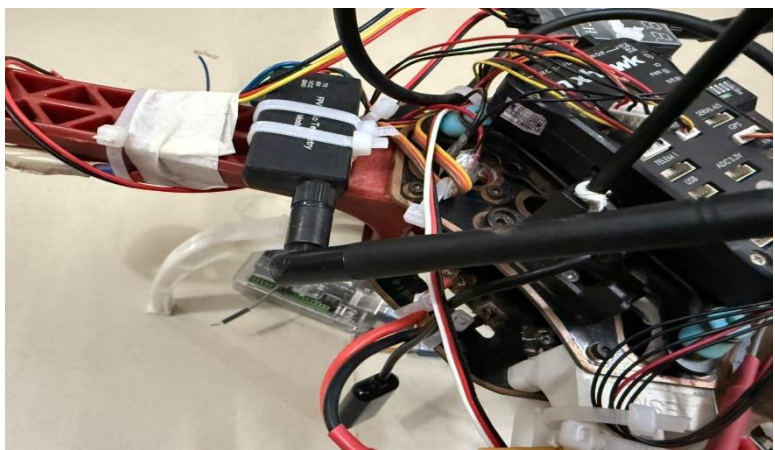
MISSION Zoom 14.0

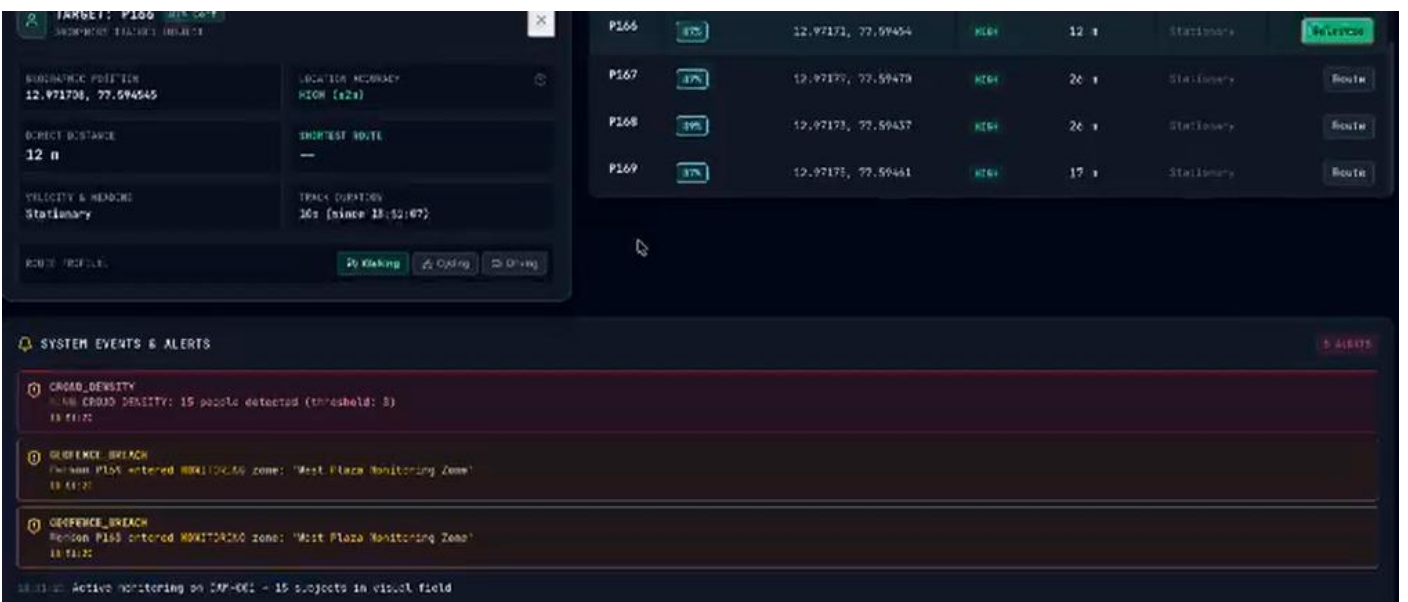
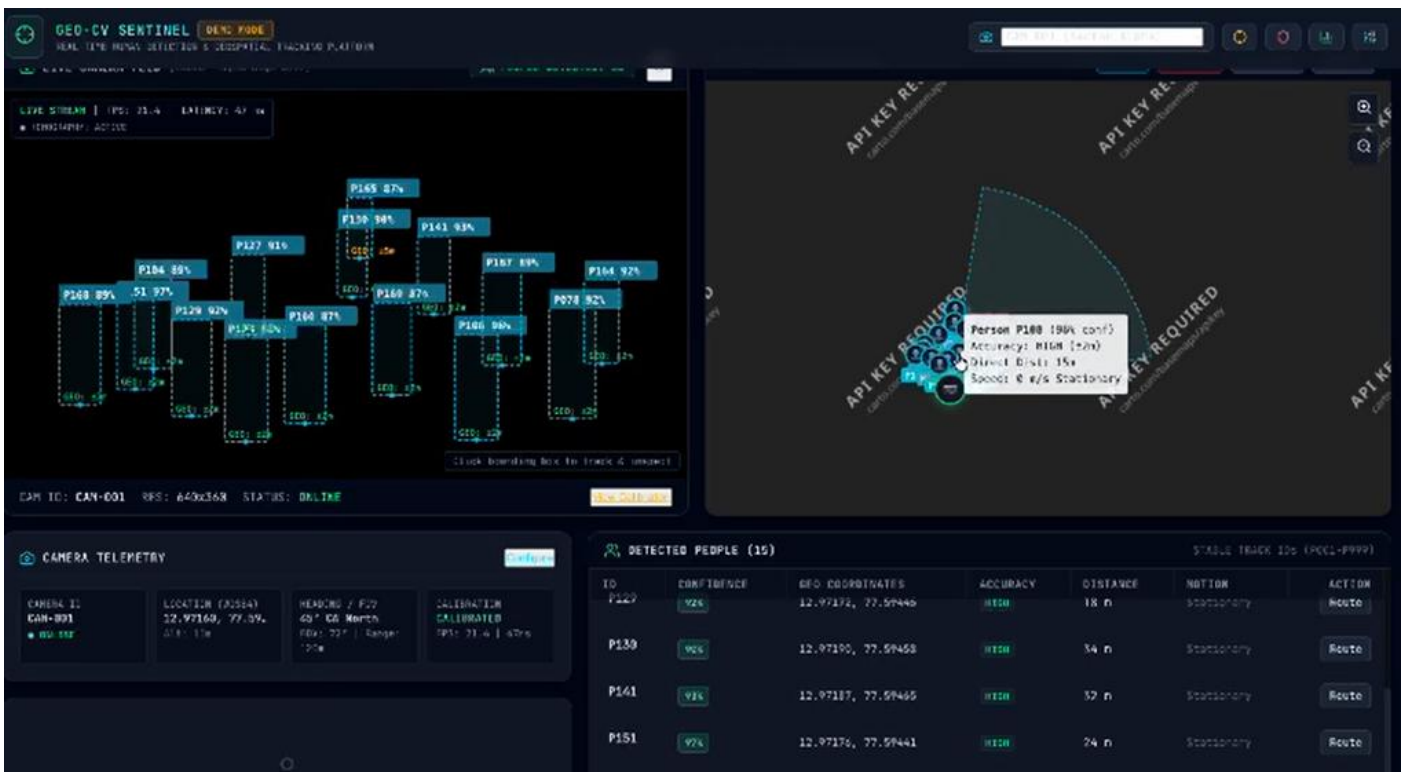
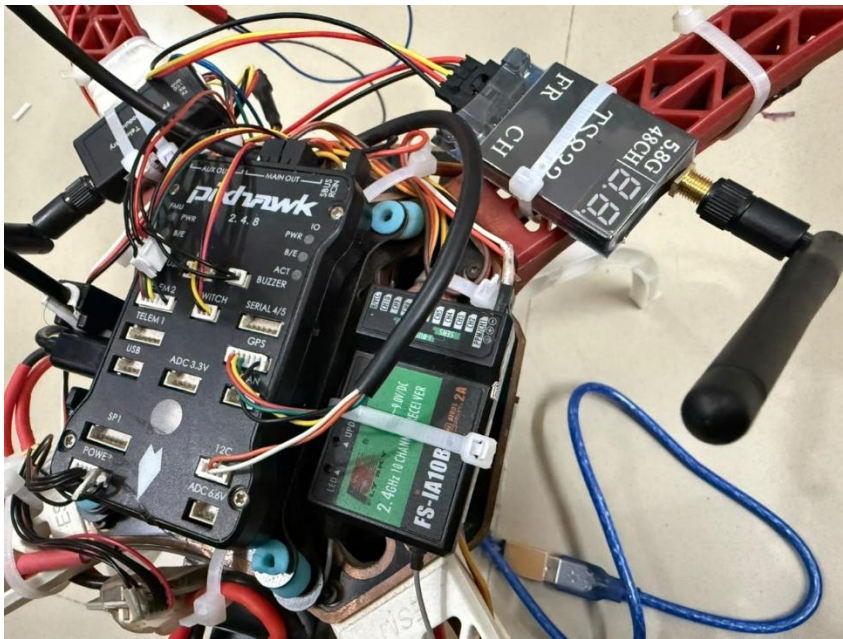
Grid View KML
GoogleSatelliteMap
Status: loaded tiles
Inject Custom Map
Load File
Save File
Read
Write
Write Fast
Home Location
Lat 12.76111961
Long 77.70393848
ASL 903.66

Refreshing Altitude Angel access token.
You need to sign into Altitude Angel. Click here to sign in.

WP Radius Loiter Radius Default Alt 50 Verify Height Add Below Alt Warn 0 Spline MAVFTP

	Command								Frame	Delete		Grad %	Angle	Dist	AZ
1	TAKEOFF	0	0	0	0	0	0	50	Relative	X		3.1	1.8	160...	295
2	WAYPOINT	0	0	0	0	12.7718762	77.7043247	50	Relative	X		4.2	2.4	119...	2
3	WAYPOINT	0	0	0	0	12.7693650	77.7103329	50	Relative	X		0.0	0.0	708.9	113
4	RETURN_TO_LAUNCH	0	0	0	0	12.7641750	77.7114487	50	Relative	X		0.0	0.0	589.7	168





SIH JUDGE QUESTIONS AND ANSWERS

1. What is GARUDA?

GARUDA is an AI-powered autonomous search-and-rescue drone designed for disaster-affected areas. It uses RGB and thermal sensing, on-device AI, geo-tagging, disaster mapping, risk assessment, and route planning to provide actionable information to rescue teams.

2. What exact problem are you solving?

During disasters, rescue teams may not know where survivors are, what hazards are present, or which areas can be approached safely. GARUDA provides aerial intelligence before responders enter the affected area.

3. What is innovative about your solution?

Our innovation is the integration of RGB + thermal sensing, on-device AI, geo-tagging, disaster mapping, risk assessment, and risk-aware route planning into a single rescue workflow.

4. Why use both RGB and thermal cameras?

RGB provides visual information, while thermal provides heat-signature information. The two are complementary and can provide additional information for detecting human presence under challenging visual conditions.

5. Why is AI processing done on the drone?

Disaster areas may have poor or no network connectivity. On-device AI allows critical detection to happen locally with reduced dependence on the internet.

6. What happens if there is no internet?

The drone continues local sensing and AI processing. Detection results can be stored locally and synchronized with the command center when communication becomes available.

7. How does GARUDA detect survivors?

RGB and thermal data are processed by the onboard AI system to identify potential human presence. When detected, the information is associated with location data and added to the disaster map.

8. How does geo-tagging work?

When a survivor or hazard is detected, the detection is associated with the drone's available GPS and localization information. This allows the detection to be represented at a specific location on the disaster map.

9. How does your recommended route work?

Detected hazards are geo-tagged and added to the disaster map. Different areas are assigned relative planning costs, and A* uses this weighted map to generate a risk-aware recommended access route.

10. Is your route guaranteed to be safe?

No. It is a risk-aware recommended route, not a guaranteed safe route. It is based on the hazards detected and mapped by the system, while the final decision remains with trained rescue personnel.

11. What happens if GPS is unavailable?

GPS is used when available. For GPS-limited conditions, the proposed architecture can use IMU and visual information with SLAM-based localization, subject to the available sensors and validation of the implemented system.

12. What is your testing strategy?

We follow a progressive approach: Component Testing → Ground Testing → Controlled Flight Testing → Subsystem Validation → System Integration Testing → Representative Scenario Testing.

13. What are your major challenges?

Our major challenges are limited flight time, payload and power constraints, AI detection reliability, difficult environmental conditions, GPS-denied navigation, and communication loss.

14. How is GARUDA different from a normal camera drone?

A normal camera drone mainly provides aerial imagery. GARUDA processes information onboard to detect survivors and hazards, geo-tag them, map the disaster area, assess risk, and provide actionable information to rescue teams.

15. What is the core value of GARUDA?

GARUDA converts aerial sensor data into actionable disaster intelligence, helping rescue teams understand where survivors and hazards are and how the affected area can be approached before responders enter the danger zone.

16. What if the AI gives a false detection?

AI detection may produce false positives or missed detections, especially in difficult environments. We plan to reduce this through suitable training data, RGB-thermal complementary sensing, confidence thresholds, filtering, and human verification of critical detections.

17. How will you handle limited drone battery life?

We will optimize payload, power consumption, AI processing, and mission planning. The system can prioritize important detections instead of continuously transmitting all raw sensor data.

18. Can GARUDA work in a large disaster area?

The coverage depends on flight time, mission area, environmental conditions, and available communication. For larger areas, our future scope includes multi-drone coordination, where multiple drones can divide the survey area.

19. Who makes the final rescue decision?

The rescue team does. GARUDA provides detections, maps, alerts, risk information, and route recommendations. It is a decision-support system, not a replacement for trained rescue personnel.

20. What is the biggest limitation of your current prototype?

The complete system still requires progressive validation under controlled conditions. Detection accuracy, flight endurance, navigation performance, and integrated system performance should be reported only after they are measured through actual testing.